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Towards a Self-Healing Polyimide for Soft Robotic Implications

Thomas P. Dean

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The University of Leeds

Faculty of Engineering and Physical Sciences

School of Chemistry

School of Electronic and Electrical Engineering

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Stick some more acknowledgments here.

Abstract

Soft robotic devices carve out a niche by excelling in situations in which traditional rigid robots are unsuitable. However, limitations in their use arise in harsh environments. This motivates the need for high performance protective sleeves. Chemical modifications involved hydrolysis with 0.2M NaOH, achieving a relative success of 5.76%. Crosslinking attempts, however, were unsuccessful due to highly-competitive intramolecular cyclisation back to starting material. A dye-based spectrophotometric test for carboxylic acids was examined, however, poor conclusive data was obtained. The thermal imidisation conditions of polyamic acid to polyimide were lowered by 40°C when $ZnCl_2$ was added. Robotic contributions were formed through the construction of a continuum robot with the addition of a polyimide sleeve. This sleeve was shown to resist temperature changes of 200°C and improve manoeuvrability of the continuum device. It was concluded that current chemical methods are inappropriate for the addition of self-healing moieties. However, the polyimide sleeve was an effective addition to the soft robotic actuator motivating further development to improve the longevity of the device.

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Intellectual Property

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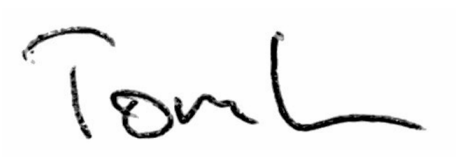

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Chapter 1

Introduction

1.1 Importance and Applications of Soft Robotics

Soft robots are devices known for their use of soft materials, compliant nature and near infinite degrees of freedom. Degrees of freedom (DOF) is a term related to the number of independent movements a robot can make in 3D space.[2] Soft robots are usually nature inspired such as "vine-like" tip growing robots,[3,4] "snake like" continuum robots[5,6] and "hand-like" soft grippers.[7,8] The advantages of using a body that is highly deformable and with near-infinite DOF means these devices can be actively controlled to navigate

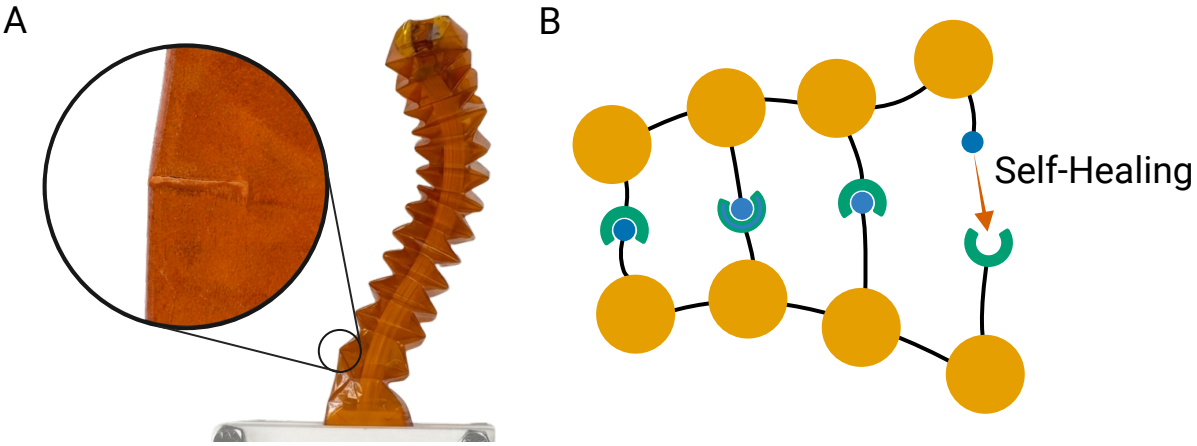

Figure 1.1: A) A continuum device after sustaining damage. B) A cartoon to demonstrate self-healing through reversible crosslink formation.

tortuous environments with ease.

This report focuses on continuum robots, often characterised by their long flexible bodies, commonly used for navigation and delivery or retrieval of objects in constrained environments.[9] Continuum robots can be classified into "continuous backbone" and "segmented back-bone" designs.[9] Continuous backbone devices are made from a soft and compliant material, in this report we will call these "soft-bodied", whereas, a segmented backbone refers to those that are made from hard, inflexible segments that are flexibly attached to achieve high curvatures, referred to here as "hard-bodied".

Hard-bodied continuum robots benefit from stiffer materials allowing a greater ease of control[9] which can improve dexterity. However, they do not easily deform and are therefore more likely to impart a greater force on the surrounding environment.[10] Hard continuum robots are often tendon driven, whereby the segments are joined by wires that can be pulled to give specific movements. Whereas soft bodied continuum robots can achieve high curvatures and low forces on surroundings,[11] but are often harder to control due to their methods of actuation[9] and can suffer from material failure.[12] Soft continuum robots commonly take the form of an inflated body, these can be controlled in many ways such as varying pressure,[3] strain limiting a specific region[13,14] or magnetic actuation.[4,10] Pressurised systems that rely on fluids need a constant seal to maintain function as opposed to a segmented tendon driven approach that can be partially damaged and remain actuatable. This motivates the utilisation of resilient materials, such as self-healing polymers, for soft robotic contexts.

1.2 Types of Self-Healing Polymers

Polymers are a type of molecule that consists of several repeating sub-units, these form chains that are combined many times over to form materials. Self-healing (S-H) polymers are a type of polymer that can heal damage.[15] This occurs through reactions and interactions between multiple polymer chains to reconnect chains that have been shifted apart

through damage. The process of damaging or cutting a polymer does not break bonds in the polymer backbone but instead can break interchain interactions called cross-linking, allowing the chains to drift apart, reforming the cross-linking interactions will "heal" the material.[16]

S-H polymers can be classified into intrinsic or extrinsic healing, and autonomous or non-autonomous.[17] Materials with an intrinsic healing capacity involve a healing mechanism through chemical bonds or interactions that are continuously present in the polymer, rather than extrinsic healing, such as capsules of "healing" material that are broken through damage and causing the material to reform.[18] Autonomy refers to the healing process occurring without external input. For example, capsules that burst upon damage of the material and release a healing agent are classified as autonomous, whereas a photochemical reaction that needs UV light to activate[19,20] is a non-autonomous mechanism.

1.3 Comparison of Different Self-Healing Pathways

There are a variety of different pathways that can be taken to create a S-H polymer depending on the chemistry of the initial polymer. The pathways that will be focused on are intrinsically self-healed types, however, both autonomous and non-autonomous types will be considered.

The three pathways below rely on a property of some electrons called conjugation. Conjugation occurs when specific electron orbitals overlap in such as way that allows the electrons to move around the molecule. This is often exploited with a property of some atoms called electronegativity, having a highly electronegative atom in a molecule will attract conjugated electrons to create a local high density region of electrons and therefore a polarisation in a molecule.

Dynamic covalent bonds can form and break based on the conditions applied to them[21] and they are ideal for use as mechanisms for S-H due to their intrinsic nature and reliabil-

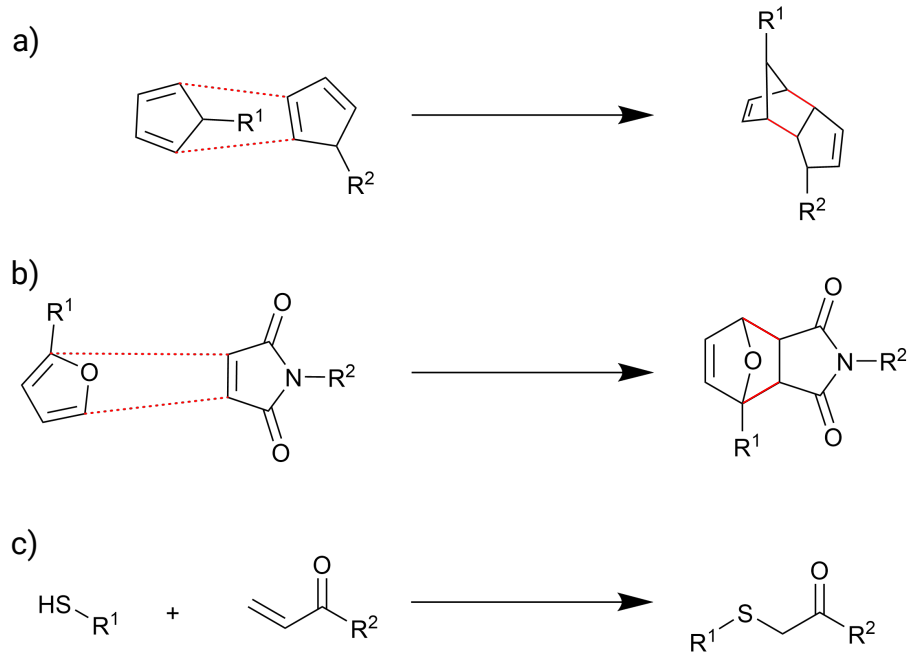

Figure 1.2: a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene.

ity in production of high yields.[22] This report investigates the use of two types of dynamic covalent bonding, Diels-Alder and Michael additions, and an example of a supramolecular interaction. Supramolecular chemistry is based on interactions that are non-covalent and may be more suited to a soft material.[23] This is due to the self-healing mechanism of the cross-links that forms interactions therefore the polymer chains can move semi-independently.[23,24] Whereas, a dynamic covalent bond has a much higher bond energy[21] and can therefore get closer to the original physical properties of a polymer.

1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway

Diels-Alder (DA) reactions require a diene, a class of molecule characterised by having two sets of double bonds, and an alkene ("dieneophile").[21,25] The reaction occurs when conjugated electrons from the diene are free to move to form new bonds with the carbons either side of the alkene double bond, this creates a new ring structure. A faster reaction is generally observed with an electron rich diene and an electron poor dienophile.[26] The

electronic character of the reagents can be altered by changing their local substituents. This class of reactions can have lots of different conditions depending on the molecules involved. Some variations can occur at room temperature,[27] which may make them an ideal choice for an autonomously healed medical device as the healing can occur whilst deployed in the body. However, many DA reactions are light[19,20] or heat activated.[28]

Some molecules such as cyclopentadiene can perform a DA reaction with itself, this has the potential to be an ideal choice for a S-H polymer as the molecule will never be in close proximity to a molecule that it cannot react with. However, some evidence has shown that the bonds formed are not very strong and therefore high tensile strength materials may not be possible.[25]

One of the most common types of DA reaction is a heat activated reaction between a derivative of both furan and maleimide.[29] They react extremely accurately and require mild conditions,[25] therefore they are very suitable for application as a robotics material. One of the reasons that this reaction is so effective is that the maleimide has electron withdrawing carbonyl groups which make the dieneophile electron poor and therefore reacting faster. However, this mechanism has a disadvantage compared to cyclopentadiene as there are two halves of the end product. Therefore, there must be two different processes occurring to create both the furan substituted and maleimide substituted chains. This increases the number of total steps in the reaction which increases the risk of low reaction yields and wasted material.

1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway

Thiol Michael additions are a type of reaction that relies on a thiol, a molecule characterised by a sulfur containing group and a conjugated alkene.[30] The conjugated alkene used in a Michael addition is with a carbonyl group that will attract the electrons from the alkene to allow the thiol group to react. This has a similar advantage to the Diels-Alder reaction that it is not totally dependant on specific reactants merely groups of

reactants. Therefore slight changes can be made to the adjoining groups to tweak the reaction conditions. This has been previously shown to have been successful with a heat activated reaction that could heal scratches after 15 to 30 minutes at 60 °C.[31]

1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway

Pi-Stacking is a type of intermolecular interaction whereby conjugated systems interact based on the neighbouring substituents determining if a region is electron rich or poor. These electron disparities cause an electrostatic interaction. This method commonly uses pyrenyl groups, these are polycyclic aromatic compounds, a very pi-electron rich species. These can be interacted with a napthalene diimide complex, an electron poor pi-system to give four pi-stacking interactions. A method has shown the creation of a polymer chain end group which can sandwich a napthalene group within two pyrenyl groups.[32] This method can be combined with other intermolecular interactions like hydrogen bonding[33] or mettalosupramolecular interactions to give a stronger interaction.[34]

1.4 Self-Healing Polymers in Soft Robotics

Continuing the theme of biological inspiration of soft robotics, S-H polymers can be used to extend the lifespan of components. In soft continuum robots, the goal is usually to make the robot softer than the environment surrounding it.[11] This presents challenges in abrasive environments that can damage the device. It is therefore evident that a damage resilient material would be useful. Damage resilient materials can take multiple forms: Damage isolation, material elasticity, physical self healing such as low melting temperature materials and chemical self healing.[35]

1.4.1 State of the Art Self-Healing Soft Robots

State-of-the-art approaches using intrinsic S-H mechanisms are commonly based on DA chemistry.[17] These have been successfully implemented in tendon-driven[36] and pneumatically controlled soft robotic devices.[37,38] Such devices are capable of recovering from macroscopic damage, including punctures and cuts. Additionally, there are non-covalent approaches that utilize hydrogen bonding in the development of soft self-healing actuators, such as in McKibben muscle designs.[39]

1.4.2 Requirements for Self-Healing Polymers in Soft Robotics

Terryn *et al.*[17] states that there are 5 main criteria for a S-H polymer; macroscopic damage must be healed completely; healing can take place multiple times at the same location; there are minimal material property changes after healing; elastomeric behaviour is achieved; the reprocessing and recycling of a material is possible. Here we propose that elastomeric behaviours are non-essential and a more useful requirement would involve a timely S-H reaction.

1.5 Limitations with Current Materials

Damage control whilst a device is active can be a challenge,[40] this shows a need for autonomous S-H.[17] Non-autonomous S-H can be made "functionally autonomous" through the use of stimuli responsive mechanisms, however, the current options are slow in operation.[40] Currently recovery timescales of applied S-H polymers are around 25 hours.[35,38] By altering the polymer used or S-H pathway this could be improved. There is also a lack of an easy synthesis route for many of the current polymer options,[35] thereby increasing the cost and reducing widespread use. Also the number of damage-repair cycles in these devices are low, such as 3 or less.[38] Most devices do not achieve all five criteria laid out

in the Terryn *et al.* review,[17] such as the McKibben design of A. Lopez Diaz *et al.*[39] only reporting a healing efficiency of 65%.

Chapter 2

Project Aims and Objectives

2.1 Aims and Objectives

The aim of this project was to investigate the use of a self-healing polyimide in a soft robotics context.

It was hypothesised that the incorporation of a S-H polyimide into soft robotic components would enhance their lifespan, increasing long term functionality and enabling real-world deployment in harsh environments.

The objectives of this project were to:

- Investigate the alteration of a polyimide through a Diels-Alder pathway to achieve a self-healing polymer
- Investigate the processability of the modified polymer through material tests such as melting point analysis.
- Based on the processability, apply the polymer to either a sheet film material or a moulded component suitable for use as a soft robotics component
- Based upon the form of the modified polymer, construct and use as an inflatable

soft continuum robot or use as hard or soft bodied components

- Test component/robot through different techniques such as tensile testing and force characterisation.
- If time allows, design and construct a system that can autonomously self heal components. For example, using a UV controlled self healing reaction, the flow rate of air into an inflatable device could be measured. Therefore meaning that if the flow rate changed a UV light could be activated, thereby autonomously self-healing the components.

This report is presented in a split-approach in order to cover the objectives in full detail. Firstly, chemical modifications are attempted and analysed in chapter 3 before robotic contributions are covered in chapter 4.

Chapter 3

Towards the Synthesis of a Self-Healing Polymer

This study aims to lay initial ground-work for the development of a self-healing polymer. This involves Kapton®, a commercially available polyimide, well known for highly resilient properties. This section covers the initial process for the alteration of polyimide via hydrolysis in order to expose reactive carboxylic acid (COOH) groups and the subsequent reactions with these exposed COOH groups. Hydrolysis is a process in which polyimide is broken down into a precursor, polyamic acid, this exposes regions in which additional reactivity can be added. One route to the development of a self healing polymer is through the formation of crosslinks at these exposed COOH sites. Specific types of crosslinks can dynamically break and reform to achieve a material that can repair itself. A proof of concept reaction between polyamic acid and ethylene carbonate, a crosslinking reagent, was attempted as a preliminary demonstration.

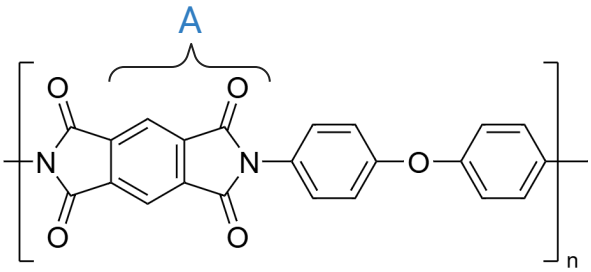

Figure 3.1: Structure of polyoxydiphenylene mellitimide (Kapton®) with higlighted pthalamide region at A

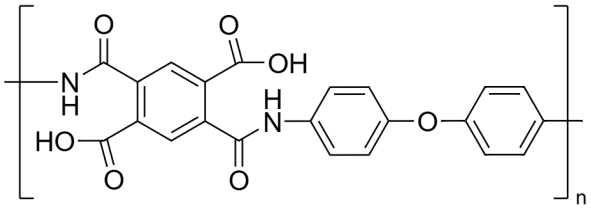

Figure 3.2: Structure of polyamic acid, the goal of hydrolysis attempts

3.1 Results and Discussion

3.1.1 Hydrolysis of the Polyimide

$$C_{22}N_2O_5H_{10} + NaOH \longrightarrow C_{22}N_2O_6H_{11}Na \tag{3.1}$$

The method for hydrolysis was based on previous literature publications[1,41–47] where an alkaline solution is used to hydrolyse polyimide to form a salt intermediate and then an acidification to form carboxylic acid groups. A NaOH solution (0.2M)[41–43] was initially trialled for the alkaline hydrolysis step at room temperature for 90 mins, however when there was minimal change under Infrared spectrometry (IR), the temperature of the reaction was increased to 120°C. Also, the NaOH concentration was adapted similar to that used in Xue (1986).[46] This comprised of the use of 1 equivalent OH^- for every 1 monomer of the polyimide. This was chosen to ensure a partial hydrolysis, to only expose one carboxylic acid group as opposed to the total hydrolysis to form a tetra-substituted benzene ring. However, during IR analysis of this reaction minimal hydrolysis was also observed.

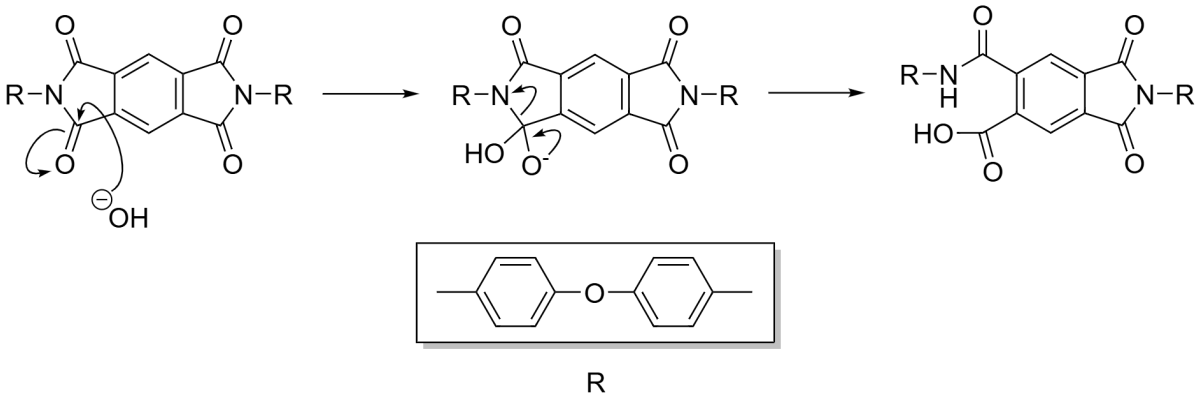

Figure 3.3: The mechanism of conversion of polyimide to polyamic acid via a nucleophilic addition of a hydroxy group

A widely accepted base-catalysed amide hydrolysis route is through a second order pathway, where a hydroxy group (OH^-), from the base, nucleophilically adds to the carbonyl group (C=O) to form a tetrahedral intermediate.[48] This cleaves into two and the amine is eliminated. In pthalamide structures (Fig 3.1A), a molecule close in character to the polyimide monomer, it has been shown to exhibit similar behaviour when in solution.[49]

However, mechanistic changes occur when the reaction contains macromolecules, it has been shown in previous literature that there is a third order dependence on the concentration of OH^-($[OH^-]$) and hence follows a mechanism that starts with the nucleophilic addition of the OH^- to form a charged intermediate before undergoing rate-determining cleavage of the C-N bond to form the associated polyamic acid.[41] Moens (1957) proposed that, due the hydrostatic repulsion, the rate of hydrolysis is lowered when a polymeric chain has increasing numbers of negative charges,[50] this would imply that the formation of the mono-anion intermediate could hinder the reaction.

While multiple literature sources opt to use KOH,[1,44,46,47] hydrolysis has been shown to occur in the presence of NaOH.[41–43] In retrospect, hydrolysis attempts using KOH should have also been approached due to the success of many KOH hydrolysis attempts.

However, most of the failure in this experiment was attrubuted to the low concentration of NaOH, as sources have stated that a concentration below 0.5M was ineffective.[1] This explained why the testing with 0.3M was also ineffective. Additionally, some of the failure

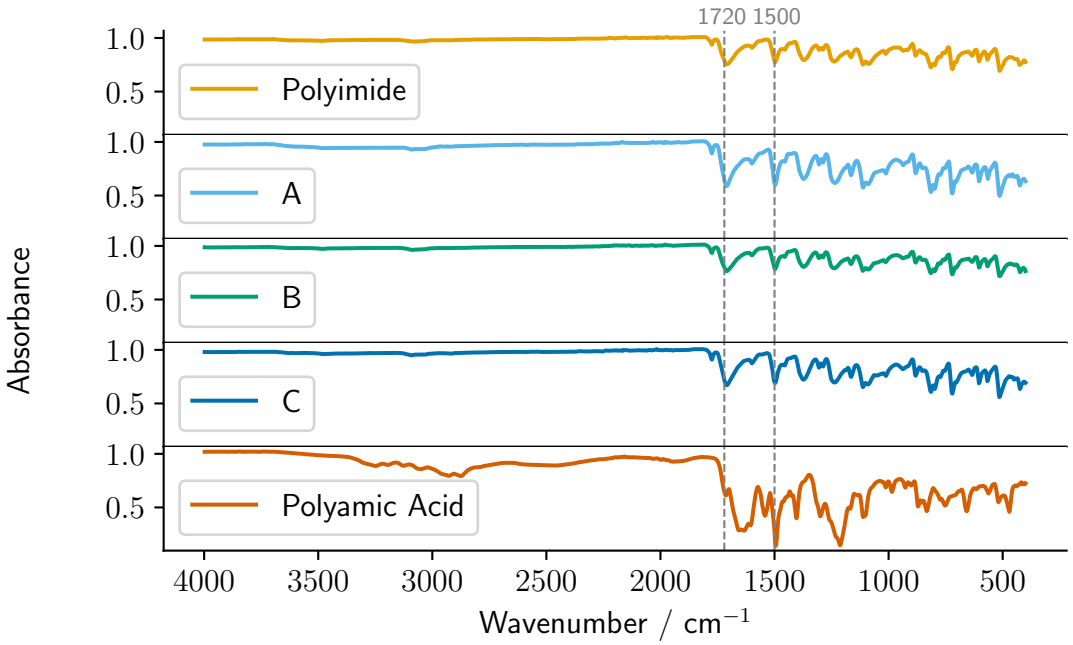

Figure 3.4: Infrared spectrometric results of the hydrolysis of polyimide to polyamic acid. Attempted with 3 conditions and concentrations: A) 90 minutes, 25°C, 0.2M NaOH. B) 20 minutes, 120°C, 0.02M NaOH. C) 40 minutes, 120°C, 0.02M NaOH.

may be caused by the lack of addition of a reagent to maintain constant ionic strength, which has been shown to assist hydrolysis with NaOH.[41]

Spectral Analysis

Structural determination was conducted via Infrared spectroscopy (IR) through comparison between the attempted hydrolysis of the sheets and a purchased form (polyamic acid).

Characteristic peaks of both the hydrolysed and the control forms were observed around 1500cm^{-1}, corresponding to the C=C bond in the central benzene ring,[1,44] along with the 1235 cm^{-1} aromatic ether stretch[44] both which do not change throughout the hydrolysis.

Also observed is the C=O asymmetric stretching mode at around 1710 cm^{-1},[1,42,44–46] the peak around 1775 cm^{-1} as the symmetric C=O stretch[44–46] and the C-N-C imide ring

Table 3.1: ATR-FTIR absorbance ratios and relative imide surface content for various samples in comparison to Wu (2017)[1]

Sample	**Base**	**Conc./M**	$\mathbf{A_{1500}}$	$\mathbf{A_{1720}}$	$\mathbf{A_{1720}/A_{1500}}$	**Rel. imide %**
90 min, 25°C	NaOH	0.2	0.2243	0.2304	1.0273	0.9424
20 min, 120°C	NaOH	0.03	0.1062	0.1167	1.0990	1.0082
40 min, 120°C	NaOH	0.03	0.1608	0.1735	1.0789	0.9898
Polyimide	—	—	0.1125	0.1226	1.0900	1.0000
PAA	—	—	0.8508	0.1735	0.2040	0.1871
Wu (2017)	KOH	0	0.0513	0.0587	1.1440	1.0000
Wu (2017)	KOH	0.1	0.0504	0.0573	1.1375	0.9943
Wu (2017)	KOH	0.5	0.0511	0.0568	1.1096	0.9699
Wu (2017)	KOH	1	0.0587	0.0617	1.0519	0.9194
Wu (2017)	KOH	5	0.0763	0.0072	0.0946	0.0827

stretch at 1370 cm^{-1},[44,45] all of which decrease in magnitude during hydrolysis.[1,45]

One literature source[51] states that the symmetric C=O stretch is at 1775 cm^{-1} and the asymmetric is at 1720 cm^{-1} however, due to the disparity between this and the other sources, this result was disregarded.

Some sources estimate their degree of hydrolysis with a comparison of the ratio of intensities of the C=O : C=C (1720:1500 cm^{-1}) peaks,[1,46] these are summarised in the table 3.1. In an unexpected shift, the symmetric C=O peak was observed slightly lower at around 1710 cm^{-1} in this report.

As in the Wu 2017 paper,[1] this report assumes the polyimide sample to have 100% imide content on the surface, hence finding that the polyamic acid sample contains 18.71% surface imide content. Also observed was the decrease in imide content relative to both an increase in NaOH concentration and time. This work found that the best hydrolysis attempt was the 90 minute, 0.2M hydrolysis at room temperature, giving a total imide surface content of 94.24%, meaning that 5.76% of the imide rings were cleaved.

3.1.2 Spectrophotometry

Due to the limitations of only using one method of analysis of the degree of hydrolysis of polyimide, an additional method was sought out. Other traditional methods of determination such as nuclear magnetic resonance (NMR) rely on solubilty of samples, this is a property which, by design, polyimide does not have. Therefore, a spectrophotometric test was proposed.

Other spectrophotometric methods to detect carboxylic acids (COOH) involve fluorescent tagging. These involve the attachment of a fluorescent label to the carboxylic acid group, producing a distinct peak in a UV-vis spectrum. Attachment reagents involved are varied, including halogenoalkyl, hydrazine, amine, alcohols and sulfonate groups.[52] However, most of these options require activation of the COOH group before binding of the fluorescent marker. This aimed to avoid any pre-activation reagents as this would lower the activation of the unwanted intramolecular amide formation, thereby reforming an imide ring.[53]

Therefore, a method proposed by Johar (1972) involving the use of Eosin Y (EY) and Methylene Blue (MB) for determination of carboxylic acids was used.[54] Dyes were mixed as according to the concentrations stated by Johar (1972). However, is was apparent that the concentration was too high and therefore additional dilutions of the dye mixtures were made. Initially tested were the interactions between the dyes and water, a peak was observed at 515nm confirmed to be corresponding to EY[55–57] and a very small peak at 680nm correspoding to MB.[58–61] The shoulder on the EY peak is formed through the dimerisation of the molecule due to the high concentration.[56]

An example of a positive test would occur with acetic acid, a simple carboxylic acid. In the spectrophotometer, the peak corresponding to the EY was destroyed. This report proposes that this destruction is due to the use of glacial (the most concentrated an acid can be) acetic acid. In high concentrations, acetic acid can to convert the EY into a precursor (dihydrofluoran)[62] which causes the observed blue shift in the peak[63] along

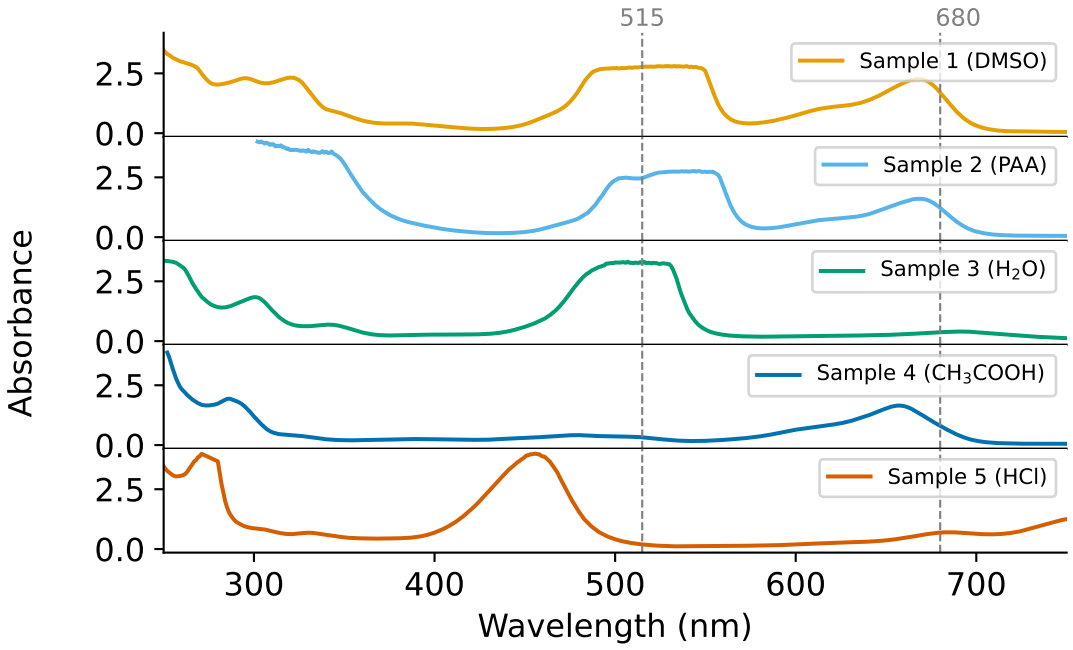

Figure 3.5: UV-vis spectrophotometric analysis of the interaction between Eosin Y, Methylene Blue and different test compounds.

with the reduction in magnitude.

Next assessed was the interaction between dye mixture and polyamic acid (PAA) dissolved in dimethyl sulfoxide (DMSO). This gave two peaks, one in the EY region at 530nm and one in the MB region at 668nm. The interaction with the MB was very similar to that of the acetic acid which was a promising result. However, in the EY region, the large peak was not similar to that of acetic acid. This report proposes that this effect was caused by EY catalysing amide formation, hence forming imide rings and removing the carboxylic acids. This implies that the peak at 530nm corresponds to the solvent (DMSO) rather than an interaction with the PAA.

To test this hypothesis, DMSO was mixed with the dyes and a peak at 517nm and 667nm was observed, which was confirmed to be EY[56] and MB.[58,60,61] As a final validation, it was checked that the interaction was not simply pH related by testing HCl, this was chosen as an acid that does not contain carboxyl groups and therefore should react differently

to the acetic acid. The absorbance spectrum showed one peak at 454nm, indicating that the interaction between HCl and the dye mix is sufficiently different to other acids that the interaction can be assumed to not be pH related.

Due to the difference in absorbance for the acetic acid and the PAA and the catalytic effects of EY on imide reformation, it was determined that the GS Johar method is not appropriate for a determination of hydrolysis of polyimide.

3.1.3 Cross-linking of the Polyimide

$$2\,C_{21}N_2O_5H_{13}{-}CO_2H + C_3O_3H_4 \xrightarrow{\text{cat}\,\cdot\,\text{salt}} (C_{21}N_2O_5H_{13})_2C_4O_4H_4 + CO_2 + H_2O \quad (3.2)$$

Cross-linking attempts follow a route whereby ethylene carbonate (EC) is used to form an ester with the carboxylic acids on neighbouring polymer chains, thereby joining them with a "cross-link". This was attempted with three different salts (Et_4NBr, $ZnCl_2$, $FeCl_3$) as catalysts.

During the cross-linking reaction with Et_4NBr, an orange-brown suspension was formed that when washed, filtered and freeze dried yielded a pale orange powder. With $ZnCl_2$, the liquid was a dark orange clear liquid that, when water was added, gave a single clump of grey polymeric residue. This was freeze dried to remove the excess water and then analysed. The cross-linking reaction with the $FeCl_3$, yielded a dark orange/brown gel, which formed a grey gum-like texture in small pieces when water was added. This was also frozen and analysed.

Safety Consideration

During a cross-linking with ethylene carbonate, CO_2 is released (Eq. 3.2). This was concerning due the sealed monowave reactor vessel which has a pressure limit of 20 bar. To ensure there was a reliable safety buffer, an artificial limit of 5 bar was set. The Ideal

Gas law was used to give the amount of EC to use to ensure this limit was held.

$$PV = nRT \rightarrow n = \frac{PV}{RT}$$

$$\text{P} = 500 \text{ kPa, V} = 10 \text{ mL, R} = 8.314 \text{ J K}^{-}1 \text{ mol}^{-}1, \text{ T} = 423\text{K}$$

$$n = \frac{500 \times 10 \times 10^{-3}}{8.314 \times 423} = 0.00142mol$$

Since the CO_2 is generated from the EC, there are same number of moles for the CO_2 and for the EC. This motivates the use of $mass = Mr \times n$, where Mr = 88, n = 0.00142

$$Mr \times n = 88 \times 0.00142 = 0.12496g$$

This result set an upper bound for the mass of EC allowed in these reactions.

Detailed mechanistic studies of the reaction between a carboxylic acid and ethylene carbonate in the presence of a quaternary ammonium salt, such as Et_4NBr, have been performed in other literature.[64–66] Yoshino (1977) states that the mechanism begins with a nucleophilic ativation by the anion of the catalytic salt on a methylene group of the EC, this cleaves the hydrogen from the alcohol group (OH) of the carboxylic acid which can now nucleophilically substitute the halide to form the ester. Fagerburg (1984) disagreed with this mechanism and proposed that the rate of the reaction was dependent on the anion of the catalytic salt. However this result was not backed-up experimentally. It was observed during a reaction of the catalytic salt and EC alone, the activation energy of the EC decomposition was much higher than in any esterification attempts. Therefore a mechanism was proposed involving the attack of a carboxylate species on the methylene carbon of the EC to decompose the ring and form the ester product. Given the depth of Fagerburg's work, this report considers this mechanism to be more robust than the one proposed by Yoshino.

The change of catalyst to a lewis acid ($ZnCl_2, FeCl_3$) was to encourage a direct activation

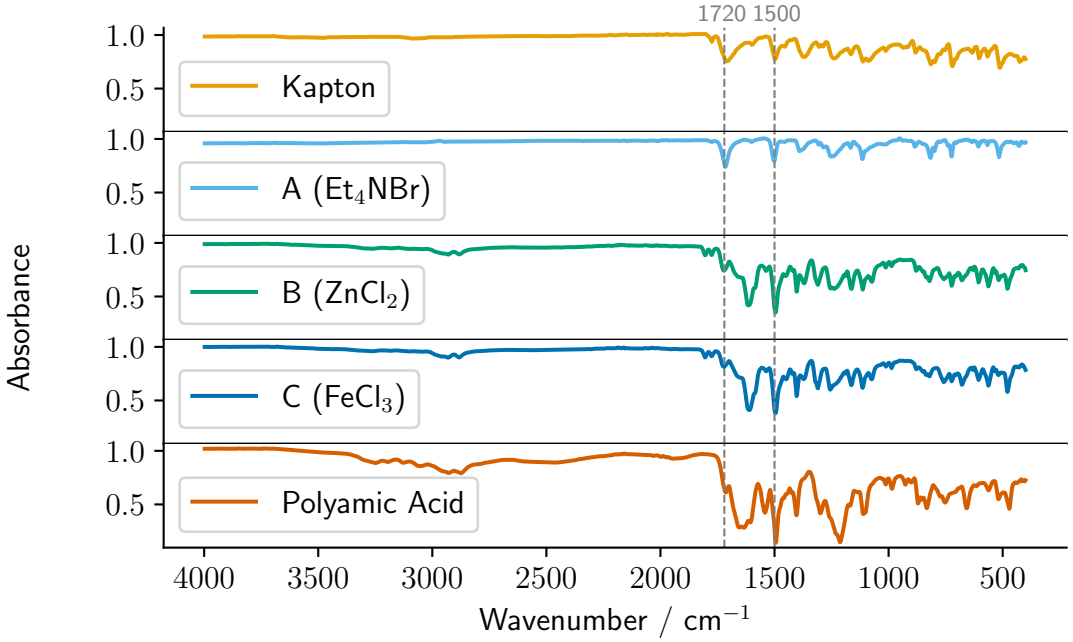

Figure 3.6: Infrared spectrometric results of the crosslinking of polyamic acid with ethylene carbonate. Attempted with 3 different catalytic salts: A) Et_4NBr B) $ZnCl_2$ C) $FeCl_3$

of the carbonyl group on the EC, the mechanism for this proposed step was as follows: The lewis acid activates the carbonyl on the EC by forming a complex. This allows the alkoxide to attack the methylene group of the EC and form the hydroxyethyl ester.[67] Von Seggern (2020) however reported, the use of zinc based catalysts was found to be ineffective in this reaction.[67]

Spectral analysis

Spectral analysis was performed mainly with IR, but also with ^1H NMR and DOSY.

From the IR data, the crosslinking with Et_4NBr was deemed to be unsuccessful as the product spectra the key peaks of the C-O bond corresponding to an ester or any alcohol formation were not observed. Alcohol peaks would be likely as crosslinking attempts are unlikely to be 100% successful, hence the single ester product would also be formed. However, neither of these were present and since the product was insoluble, no NMR could

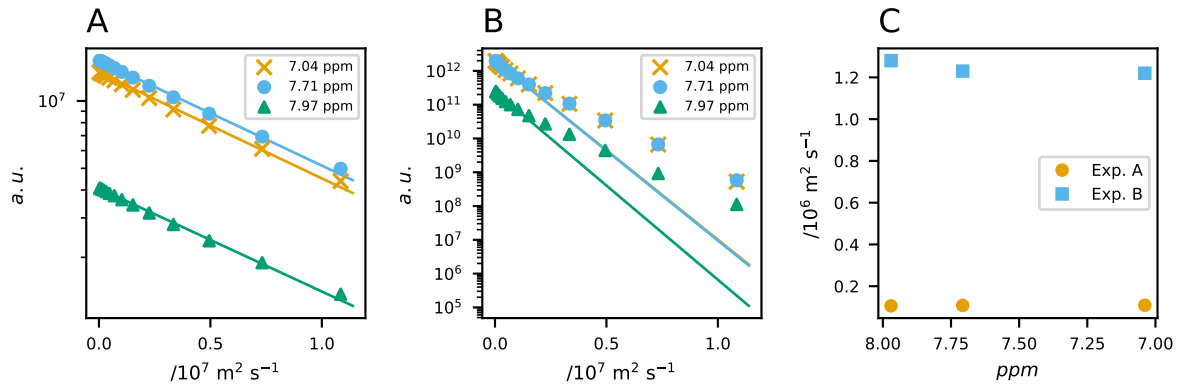

Figure 3.7: Resulting DOSY plots from the catalytic crosslinking of polyamic acid with ethylene carbonate and $ZnCl_2$. A) Polyamic acid in DMSO. B)Polyamic acid after reaction with Ethylene carbonate and $ZnCl_2$ for 1 hour at 120°C. C) Relative speed of diffusion of bulk group peaks for Sample A versus Sample B.

be obtained. This IR spectrum was compared with the spectrum of polyimde (i.e. the ring closed form of the polyamic acid) and the similarities confirmed that the imide ring reformation had occured. Key spectral peaks confirming the lack of crosslinking contain the C-N-C peak, commonly found at 1370 cm^{-1}, however in our sample we observed this peak at 1391 cm^{-1}. Other signs of imidisation include the increase in the magnitude of the C=O stretch at 1715 cm^{-1}. These peaks would indicate the imide ring reformation as opposed to the esterification product.

From the IR data of the $ZnCl_2$ and the $FeCl_3$ attempts, we observe two new peaks at around 1800 cm^{-1} which are not in either the polyimide or the PAA samples. This shows the presence of EC in the final sample as C=O bonds in liquid EC[68] are found in this regon. Since these samples were soluble, proton nuclear magnetic resonance (^{1}H NMR) was available. A new peak was observed at 4.43ppm in the ^{1}H NMR spectrum, however, when adding 2 μL of EC in a "spike test", the peak increased in size confirming that the new peak was starting material (EC). Other peaks in the $ZnCl_2$ and $FeCl_3$ samples are inconclusive due to their presence in both samples (aromatic C-H peaks). After increasing the temperature of the $ZnCl_2$ catalysed sample, the IR peak positions show the intramolecular cyclisation of the imide ring with the C-N-C stretch at 1370cm^{-1}.

Diffusion ordered spectroscopy data (DOSY) was collected for the PAA sample and the

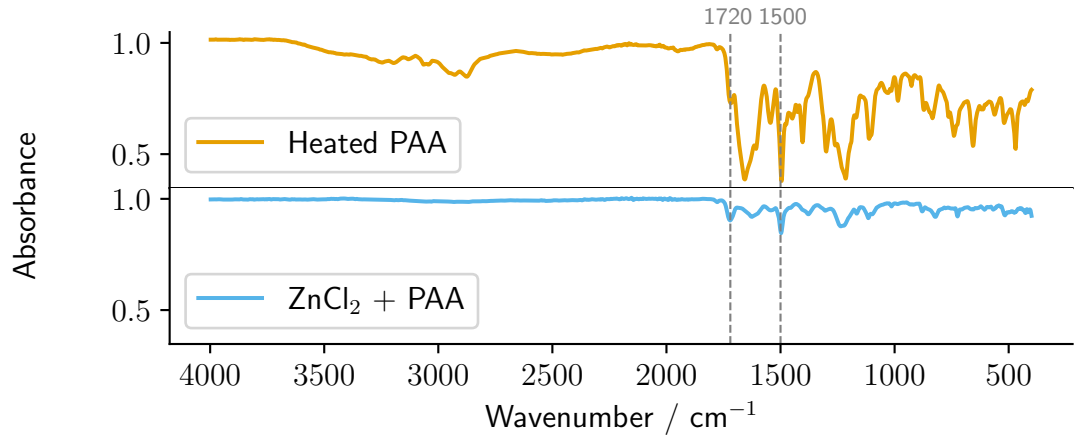

Figure 3.8: Infrared spectrometric results of the reduction in thermal imidisation conditions of polyamic acid to polyimide. Two samples were heated to 110°C for 1 hour: polyamic acid versus polyamic acid and $ZnCl_2$.

$ZnCl_2$ sample, these are shown in fig. 3.7 as A0202 and B0202 respectively. Single parameter diffusion models were fitted to the integrals of bulk group peaks (7.04, 7.71 and 7.94 ppm). In the case of PAA, good fitment can be observed with a mono-parameter exponential ($y = B \times e^{-Fx}$) (where F is the decay rate constant and B is the amplitude) model unlike with $ZnCl_2$ where the linearised exponential line is steeper than the clear trend-line of the data. For a better fitment of the data, a multi-parameter model could be calculated. However for this level of analysis, no further calculations are neccessary.

The speed data plot, plots the decay rate constant of A0202 versus B0202. From this it can be observed that B0202 diffuses significantly faster than A0202. With this limited data an assumption was made that the polymer chain length in B0202 has shortnened as compared to A0202.

3.1.4 Catalytic Imidisation of polyimide

During the attempted crosslinking reactions, the imide ring formation was observed at a lower temperature than previously seen in the literature.[69] This motivated attempts to recreate this with only polyamic acid (PAA) and $ZnCl_2$.

Without the catalyst, material became slightly darker after heating for an hour. This

Table 3.2: IR peak positions of thermal imidisation of polyamic acid to polyimide, peak absorbances, and relative imide surface content to polyimide.

Sample	ν_{1500} **/cm**$^{-1}$	$\mathbf{A}_{1500}$	ν_{1720} **/cm**$^{-1}$	$\mathbf{A}_{1720}$	$\mathbf{A}_{1720}/\mathbf{A}_{1500}$	**Rel. imide %**
PAA	1495.02	0.8508	1709.18	0.1735	0.2040	0.1871
Heated PAA	1495.02	0.4209	1717.34	0.1375	0.3268	0.2998
With $ZnCl_2$	1497.06	0.0727	1721.42	0.0443	0.6084	0.5581
Polyimide	1497.06	0.1125	1709.18	0.1226	1.0900	1.0000

was attributed to some minor solvent evaporation and therefore, substrate becoming more concentrated. A similar colouration after heating the sample containing $ZnCl_2$ was observed, however, there was also material suspended. This was proposed as indicative of a partially successful attempt.

An assessment of the success of the imidisation can be made by comparing the A_{1720} to the A_{1500} in the same manner as previous analysis.[1] In table 3.2, again the polyimide sheet was assumed to have 100% surface imide content and others were calculated relative to this. This report finds that the imide content of the polyamic acid heated without the catalyst had 29.98% imide content as compared to the baseline value from the unheated PAA at 18.71%. However, the sample with $ZnCl_2$ catalyst presents as successful as 55.81% imide content is observed. Also observed was a shift in peak position in the 1500 peak when imide content is over 50%. To confirm the degree of imidisation in the future, a pure sample could be obtained through a recrystallisation from DMSO.

3.2 Experimental Methods

3.2.1 Equipment

A MonoWave reactor (MonoWave 50 Closed Reaction Vessel; ANTON PAAR) was used for the majority of experiments covered in this report. This reaction vessel contains integrated temperature and pressure control and a magnetic stir element. This allowed the user to perform reactions more quickly as samples could be heated above their ambi-

ent pressure boiling points. The MonoWave reactor required a specific 10mL tube that was sealed with a rubber cap and a PTFE septum, to ensure agitation, an oval-shaped magnetic stir bar was included.

3.2.2 The Hydrolysis of a Polyimide

Polyoxydiphenylene mellitimide (Kapton®) film (0.18g, thickness: 0.05mm) was cut into pieces of approx. $3mm^2$ and placed into a beaker. This was stirred at room temperature with a solution of NaOH in DMSO (0.3M, 22ml). There was no observable difference and an Infrared (IR) spectrum was taken.

Another sample of Kapton® (polyimide) film was cut into approx. $3mm^2$ pieces and placed into a MonoWave reaction tube along with a solution of NaOH (4.2ml, 0.02M). This tube was placed into a MonoWave reactor that was set to ramp for 20 minutes and hold for 20 minutes at 100°C. There was no observable difference, therefore sample was taken and the remaining mixture was placed onto the MonoWave under the same conditions. The mixture was filtered and thoroughly washed with water before IR was collected.

3.2.3 Spectrophotometry as a Determination Method for Hydrolysis

The preparation of samples for the spectrophotometric analysis was as follows:

3 mL 0.5% Eosin Y solution was added to 1.5mL 0.1% Methylene Blue solution. This solution was diluted with 10 parts water and 5 parts water (where a "part" is 1.5mL) to give three dye mixes (summarised in table 3.3)

Test compound (2.7mL) was added to a sample vial along with one of the dye mixes (0.3 mL) to form samples summarised in table 3.4. These were inverted to ensure mixing

Table 3.3: Dye mixtures for examination in a spectrophotometer with dilution by number of parts water. Here a "part" is defined as 1.5mL.

Dye Mix	Parts H_2O	Parts EY (0.5%)	Parts MB (0.1%)
Mix 1	10	2	1
Mix 2	5	2	1
Mix 3	0	2	1

Table 3.4: Samples for testing in a spectrophotometer using a Eosin Y and Methylene Blue dye mixture.

Sample number	Test compound	Dye mix
S01	Dimethyl Sulfoxide	Dye 1
S02	Polyamic Acid solution	Dye 1
S03	Water	Dye 1
S04	Acetic Acid	Dye 1
S05	Hydrochloric Acid	Dye 1
S06	Dimethyl Sulfoxide	Dye 2
S07	Polyamic Acid solution	Dye 2
S08	Water	Dye 2
S09	Acetic Acid	Dye 2
S10	Hydrochloric Acid	Dye 2
S11	Dimethyl Sulfoxide	Dye 3
S12	Polyamic Acid Solution	Dye 3
S13	Water	Dye 3
S14	Acetic Acid	Dye 3
S15	Hydrochloric Acid	Dye 3

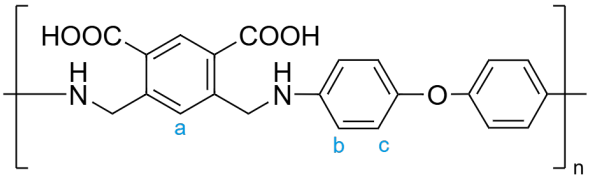

Figure 3.9: The structure of polyamic acid with bulk group hydrogen atoms marked. Due to the symmetery in the molecule some labels have been ommitted.

Table 3.5: Catalysts, reaction temperatures and analytical methods used in crosslinking attempts of polyamic acid with ethylene carbonate.

Catalytic Salt	Mass Salt /g	Reaction Temp. /°C	Analytical Method
Et_4NBr	0.08	150	IR
$FeCl_3$	0.02	100	IR, NMR
$ZnCl_2$	0.03	100	IR, NMR, DOSY

before being transferred to a plastic cuvette (BRAND UV-Cuvettes (PMMA)) and placed in a spectrophotometer (Agilent Cary 3500) to test between 250 - 750nm.

3.2.4 Crosslinking of Polyamic acid

This procedure outlines the crosslinking procedure of polyamic acid (PAA) via ethylene carbonate (EC) using a catalytic salt as outlined in previously published procedures.

A mixture of polyamic acid (0.51mL, 15.0-16.0 wt. % in NMP) was placed into a MonoWave tube with ethylene carbonate (0.08g) and a catalytic salt. This was placed into the MonoWave to ramp for 20 minutes and hold at the reaction temperature for an hour. The products were washed with water and filtered before being freeze dried and examined on the spectrometers. The reagent amounts and processing methods are summarised in table 3.5.

Tetraethylammonium Bromide (Et_4NBr) Crosslinking Spectral Data

A orange powder (0.07g, 0.171mmol) was isolated, ν max/cm^{-1}(solid); 1715, 1601, 1501, 1391, 1307, 1252, 1167, 1116, 885, 815, 800, 724 634 606, 567, 516, 428

Iron Chloride ($FeCl_3$) Crosslinking Spectral Data

A white gum (0.127g, 0.31mmol) was isolated, δ H(400MHz, DMSO); (2H, d, Ha) 10.53, (4H, s, Hb) 7.70, (4H, m, Hc) 7.03, ν max/cm^{-1}(solid); 2931, 1805, 1774, 1721, 1609, 1497, 1403, 1372, 1309, 1256, 1163, 1114, 1075, 987, 822, 759, 724, 679, 606, 563, 479

Zinc Chloride ($ZnCl_2$) Crosslinking at 100°C Spectral Data

A grey gum (0.063g, 0.151mmol) was isolated, δ H(400MHz, DMSO); (2H, d, Ha) 10.55, (4H, m, Hb) 7.69, (4H, m, Hc) 7.14, ν max/cm^{-1}(solid); 3263, 3112, 1679, 1603, 1583, 1530, 1493, 1403, 1367, 1316, 1263, 1187, 1163, 1081, 1036, 1008, 836, 722, 706, 630, 600, 559, 534, 502, 404

3.2.5 Catalytic Imidisation of Polyamic Acid

Polyamic acid (0.51mL), $ZnCl_2$ (0.02g) and a magnetic stirbar were added to a MonoWave tube. This was placed in the MonoWave to ramp for 20 minutes and hold at 110° for 60 minutes. A control experiment without the $ZnCl_2$ was also placed under the same conditions. The control experiment gave a clear dark brown viscous liquid which, when dried, gave a gum-like pale orange solid. The test with the $ZnCl_2$ gave a similar dark brown liquid, however this was slightly cloudy and, when processed, gave an orange powder.

Control Spectral Data

A grey gum (0.057g, 0.1389 mmol) was isolated, ν max/cm^{-1}(solid); 3196, 2876, 1952, 1717, 1656, 1544, 1495, 1403, 1299, 1216, 1112, 1028, 985, 926, 871, 834, 740, 657, 561, 520, 469

Catalytic Imidisation Spectral Data

A grey gum (0.076g, 0.1852 mmol) was isolated, ν max/cm^{-1}(solid); 2880, 2146, 1721, 1626, 1542, 1497, 1379, 1303, 1236, 1167, 1116, 1014, 879, 822, 724, 655, 608, 567, 516, 426, 416

Chapter 4

Construction of a Folded Actuator

In this chapter, the focus shifts towards robotics. Here a device is constructed in which the majority of the outer surface consists of the same polyimide in which chemical modification was previously attempted. This section aims to make use of the highly resilient properties that are already present within the polyimide material and demostrate the capabilities of a device constructed with them.

This chapter covers the design and construction of a continuum robot with a protective outer layer constructed from folded polyimide sheets. This protective layer is tested for temperature stability and the manoeuvrability of this device is also tested in 3D space. The device is then demonstrated in an arbitrary setting and a medical setting, a space that can require resilient materials.

4.1 Design

Actuators were constructed using a combination of a folded polyimide sleeve and an internal multi-lumen silicone tube with nitinol wires for actuation (see fig. 4.1 A). The design of this device was inspired by similar origami based approaches to construct continuum devices.[70–72]

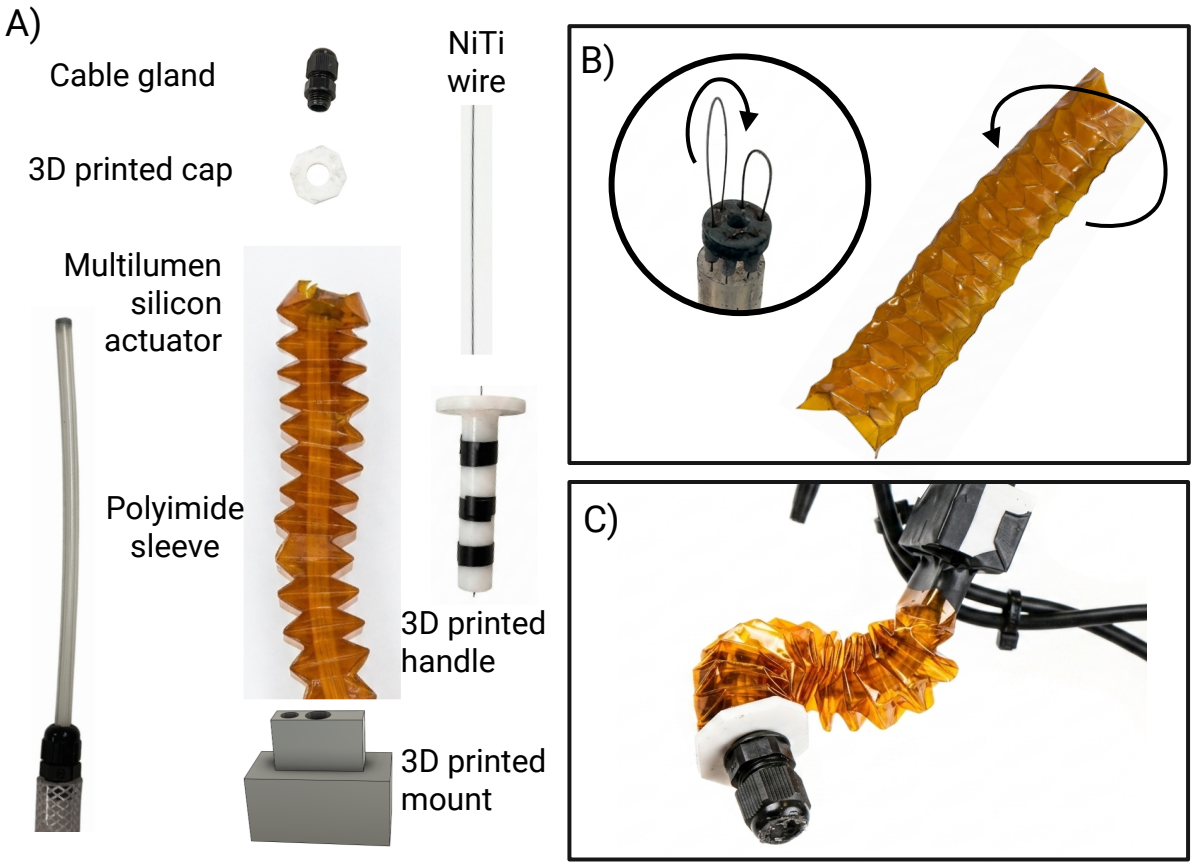

Figure 4.1: A) The components required for the construction of a polyimide sleeved, tendon driven actuator. B) NiTi wires are threaded through the top of the multi-lumen silicon tube before being threaded through an adjacent lumen. Polyimide is folded before being rolled into a protective origami sleeve. C) The completed assembly.

4.1.1 Construction of a Folded Protective Sleeve

Protective sleeves are a common addition to soft actuators in order to protect against harsh environments such as abrasive surfaces, high heat, strong chemicals and UV radiation. Nanomaterial based coatings are often at the forefront of resistance mechanisms with some focused on flame and heat resistance and others for chemical resistance through superomniphobia[73] (repellent to almost all liquids). However nanomaterials are only suitable as coatings rather than the main building material, therefore the use of high performance resilient polymers has become more popular.

In this report, we use polyimide (Kapton®, thickness: 0.05mm, Goodfellow) to form the main body of the protective polymer sleeve. Polyimides have been used before in protective sleeves and have been shown to be resistant to harsh environments. Possibly

the most famous example of a polyimide as a protective layer was on the surface of Eagle, the Apollo 11 lunar landing module. This was coated in Kapton® film due to its thermal insulation properties. One of the harshest environments that robotics faces is space, the materials chosen are lightweight, highly compressible, extremely temperature stabile and highly UV radiation resistant.[74] Other applications, such as in medical devices, require chemical resistance, particularly useful when presented with the low pHs of stomach acid (pH 2-3). Kapton® film has been shown to be highly resilient to UV and heat,[69,74] highly compressible when folded into cylinders[74] and relatively resistant to acids. Fuming nitric acid (HNO_3) has been shown to degrade polyimide,[69] however this was due to the highly oxidising nature of HNO_3. This is unlike hydrochloric acid (the main component of gastric acid) which is not an oxidising agent. In addition to this, Kapton® and other polyimides contain mellitimide ring structures allowing the partial break down of the polymer chain thereby motivating the addition of intrinsic functionality such as self-healing Diels-Alder groups without the complete conversion back to oligomer material.

There are three main designs of folded cylinders used in soft robotics: Kresling, Yoshimura and Accordion. Possibly the most popular design is Kresling, which was discovered through the twist buckling of paper cyclinders,[75] this was adapted to form consistent folds through the segmentation of a flat sheet into triangles to form a helical folding pattern.[76] Yoshimura patterns are formed through the segmentation of a sheet into parallelograms in a diamond configuration, these parallelograms are bisected with horizontal lines to form triangles.[77] Yoshimura designs can be modified through the inversion of specific protruding vertices to form a pattern known as Accordion. This transforms the parallelograms into hexagons and therefore the triangles into trapeziums.

An in depth analysis of the Accordion versus Kresling designs was performed by Butler (2016),[74] here it was found that the number of folds to make a functional device was approximately 10× less in the Kresling design than the Accordion. Often the initial failure of devices occured on the vertices, such failure was observed to occur less in Kresling designs than in Accordion. Also, when comparing compressibility (using Eq.

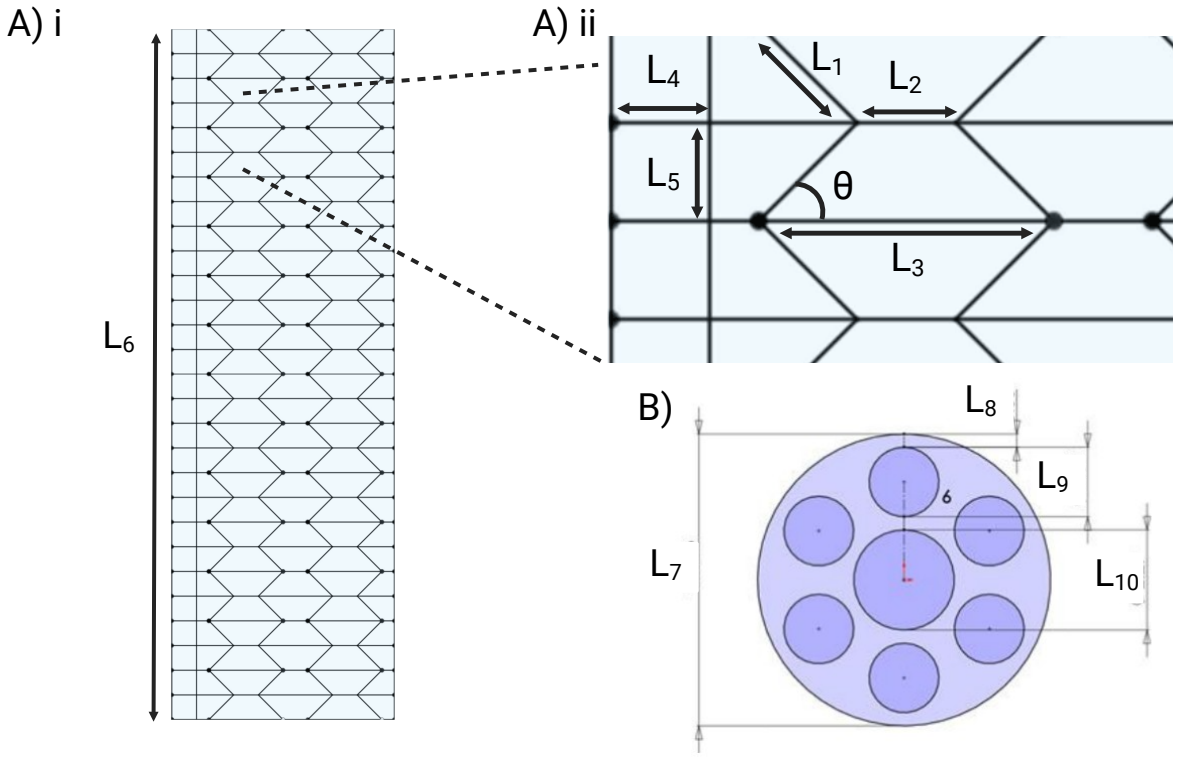

Figure 4.2: A) Design parameters of the polyimide folding pattern. B) Extrusion design parameters of the multi-lumen silicone actuation device.

4.1) it was found that, while both designs have a compressibility over 90%, the Kresling design slightly outperforms the Accordion by 1.5%.

$$compressibility = \frac{h_{deployed} - h_{compressed}}{h_{deployed}} \tag{4.1}$$

Due to the configuration of the nitinol wires in the multi-lumen silicon actuation device, it was observed that the controllability was improved when the device had torsional stability. While there are methods of adapting a Kresling design to limit torsional movement through the reflection of alternating layers. These devices are inherently prone to torque due to the helical design. Accordion Yoshimura designs are significantly more torque resistant in comparison. This is one of the primary reasons for the use of the Accordion folding pattern in this work. The design parameters are summarised in figure 4.2 and table 4.1.

Table 4.1: Design parameters for the construction of the folded polyimide sleeve

Symbol	Value
L_1	14.14 mm
L_2	10 mm
L_3	30 mm
L_4	10 mm
L_5	10 mm
L_6	280 mm
L_7	6.66 mm
L_8	0.3 mm
L_9	1.59 mm
L_{10}	2.29 mm
θ	45°

Manufacturing of a Folded Polyimide Sleeve

To manufacture an Accordion patterned sleeve, a 2D drawing to map the fold lines was drawn in Fusion360 (Autodesk) and printed onto paper. This paper was taped to a table and a sheet of polyimide was taped over the top, the polyimide is clear so the printed lines were visible through. The lines were then traced onto the polyimide with a ruler and a ballpoint pen. The ballpoint pen marked the desired fold lines and also partially indented on the lines. The sheet of polyimide was removed from the table and the outer edge was cut with scissors. The polyimide sheet was then folded along the lines as shown in fig. 4.2. The folds on the sheet were inverted and the polyimide was rolled into a cylinder. This was taped along the overlapped edge to form a temporary seal while maintaining care to line up the folds. The tape was removed in small sections and JBweld HighHeat (Farnell) was applied until the whole adhesion region. This was threaded onto a plastic tube with a diameter approx. 2mm smaller than the polyimide sleeve. The sleeve was taped to the tube and the adhesion region was taped over the top to clamp the glued region. This was left for 24 hours to cure. The glueing and the curing process was performed in a fume cupboard due to the risk of respiratory irritation through the curing process. After curing, the sleeve was removed from the plastic tube and a strip of polyimide tape was placed over the edge on the adhesion region. This improved the flexibility of the region.

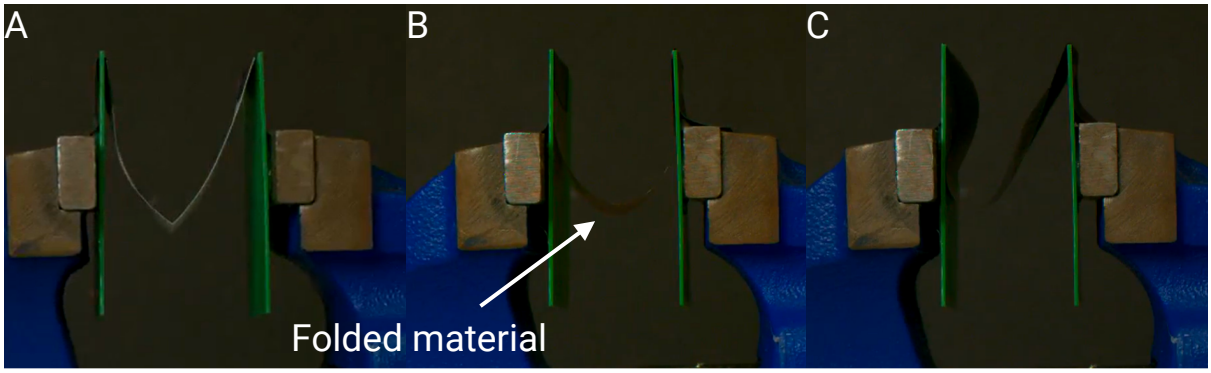

Figure 4.3: Controlled folding of sheet materials after scoring with a plotter. A) 80gsm paper, B) Polyimide thickness : 0.05mm C) Polyimide thickness : 0.025mm

One of the ends was folded over to make a cap and this was also glued. In this work, this assembly is known as the "Kapton® sleeve" or "polyimide sleeve".

Manual folding versus using a plotter

A commercially available plotter (Cricut®) was used with a scribe tool and a cutter to achieve a rapid manufacturing of a folded sleeve. This was attempted with paper initially for cheap prototyping and then with the polymer film. This would avoid the time consuming folding steps

The plotter was set up with the Cricut® Design Space software. This software allows the user to draw and communicate with the plotter via cable. The plotter has an adhesive cutting mat to hold the material in place whilst the tool interacts with it. For our testing, a rectangle (10 x 5 cm) was drawn and labelled "cut" and a line (5cm) was drawn to split the long edge into two squares (5cm^2), this line was set to "score". These two drawings were attached so as they would be performed in the same material loading cycle, the scoring was set to be first in the Cricut Design Space "job queue" to avoid the potential for the score tool to catch the newly cut edge of the polymer and cause the adhesion between the sheet and the polymer to fail. A piece of the desired material was adhered to the cutting mat and the plotter was started. This method was repeated for 2 thicknesses of Kapton® (0.05mm and 0.025mm) and 80gsm paper (roughly 0.25mm thick).

The three pieces of prepared material were taped by the top edge to the jaws of a vice (fig. 4.3). The jaws of the vice were enlarged by 3D printed pieces to ensure that the clamping would occur across the whole of the material. The vice was manually controlled to compress the pieces of material into the crease to form a fold and then slowly decompressed to evaluate the fold. The folding with the paper was highly effective, when the jaws were opened the paper was clearly separated into two sections by a fold in the centre. Both thicknesses of polyimide were not effective, when they were decompressed there was no clear fold line and the material was left hanging in a curve. The plotter can visible score the polyimide, however, the score lines do not define a crease as the scoring tool does with paper. This phenomenon is explored further by Dharmadasa (2020),[78] where polyimide is shown to not fold precisely.

It was therefore determined that the use of a plotter could be useful in the rapid production of folded sleeves, however when using this particular material, the plotter is not a viable route for creasing polyimide. A compromise could be to use a plotter to draw the fold lines by using a pen in place of the scribe tool, allowing user to bypass the printing and tracing the pattern by hand.

Limitations with the design and manufacturing

The actuator exhibits self-deploying characteristics, which are present in other devices. This is a product of treating the material as having non-zero thickness.[79] Steps can be taken to avoid this by adjusting the location of the folds. However when folding by hand, there is a high degree of imprecision that moving the folds by the thickness of the width of a sheet is not necessary.[80]

Previously, the achievement of an actuator with "wrinkle-free" compression, the triangles in Yoshimura designs should be right angled isosceles[81] and hence, the hexagons in Accordion should have a horizontal short side length equal to $\sqrt{2}a$ and long side length to be $2\sqrt{2}a$ if the angled sides are of length a. The design in this work does not feature

wholly right angle triangles as the hexagonal segments can be split into four right angled isosceles triangles and two squares. However, the slight deviations from perfect compressibility in this device have been attributed to the variability in the folding pattern rather than the optimised design mentioned by Suh (2021).[81]

The polyimide was observed cracking on the vertices, although initially thought to be repairable with a patch of polyimide tape, the complexity of bending in that region meant consistent adhesion was not achieved. This meant the constructed sleeve was not air tight and therefore locomotion via changing air pressure was not successful with the current design.

4.1.2 Complete Assembly

The complete assembly was formed with the polyimide sleeve, the multi-lumen silicone continuum device and 3D printed parts.

The silicone continuum device contains a silicone 7-hole multilumen tube with a 3D printed cap, the lumen are arranged so that there are 6 equally spaced edge lumen and one central lumen. There are 3 nitinol wires (NiTi, 0.4mm) that are approximately double the length of the silicone tube which are threaded through one of the edge lumen, through the 3D printed cap and back through a neighbouring lumen. This is repeated another 2 times until all 6 outer lumen contain a nitinol wire. At the proximal end, these wires are fed into three 3D printed handles and clamped with 1.5mm grub screws. This assembly is threaded through a stiffer tube (approx. 15mm diameter) and a cable gland on the end of this stiff tube holds the silicone tube and the larger tube together. The device can be actuated by fixing the stiff tube in place and pulling on the 3D printed handles. In some instances, it was desirable to increase the stiffness of the device by inserting a nitinol wire (0.6mm) through the central lumen. This centre lumen can also be used for adding to the functionality of the device by presenting an opportunity for targeted drug delivery, a tool channel or for connecting wires from tip-based components.

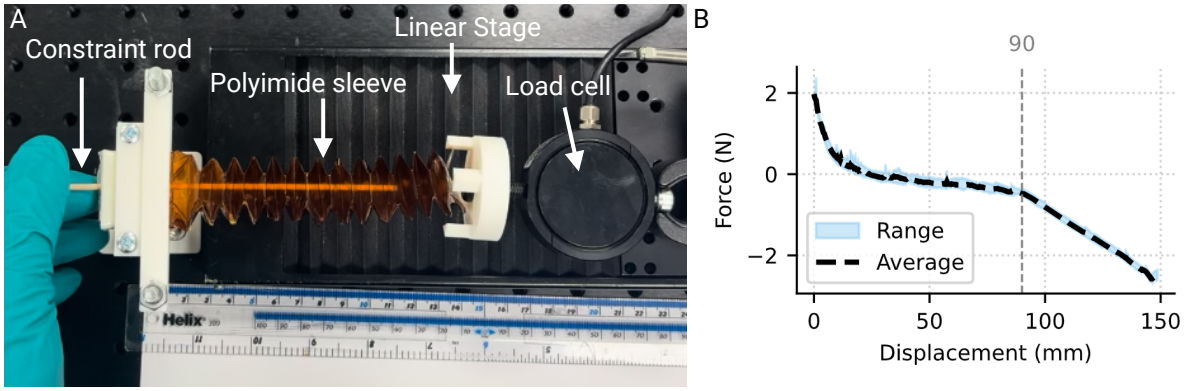

Figure 4.4: A) Experimental rig for stiffness characterisation. B) Resultant plot from the uniaxial compression/tension test of an Accordion style folded polyimide sleeve.

The construction of the full assembly is as follows. The multi-lumen silicone continuum device is threaded through the proximal clamp. The polyimide sleeve is placed around the silicone device and attached to the proximal clamp. The distal end of the silicone device is threaded through a 3D printed octagonal cap and through a PG7 cable gland. The cable gland and octagonal cap are glued to the polyimide sleeve and the gland is fastened around the silicone tube.

4.2 Results

4.2.1 Stiffness Characterisation

During compression of the polyimide sleeve, it was observed that it acted as a non-linear spring with a maximum force (shown in figure 4.4) of 2.4N at a 90mm displacement from the relaxed state. This is consistent with literature expectations for a uniaxial loading of origami cylinders.[82] Reid (2017) compared the forces applied to different origami cylinder designs using acetate (thickness: 0.1mm) that had been perforated with a laser to create folded regions. Reid (2017) also observed that the compression of Accordion cylinders was non-linear and extremely unpredictable, however the design used was not optimally designed and therefore largely incompressible.[79]

Extension was observed as linear with the maxium force of 2.6N with 60mm displace-

ment relative to the relaxed length. The relaxed length of the sleeve was 120mm, the compressed length was 30mm which therefore meant that extension of the sleeve began at 90mm displacement. The extension of the device was measured to 150mm as this was the full travel of the linear stage. It was observed that around 145mm, some vertices on the sleeve inverted and therefore, if compressed, manual reversal of these vertices was needed. In future work, the linear stage should be configured to travel in both directions to allow a full hysteresis profile to be examined.

Due to the compression and extension of the sleeve being collected on the same experimental rig, the load cell was tared at the relaxed length (120mm length, 90mm displacement) of the sleeve meaning that compression was measured as positive and extension as negative. However, when the data was collected it was observed that the force became zero at only 20mm displacement. This behaviour was attributed to the polyimide displaying an elastic characteristic where the folds settled into a more compressed form.[78] This settling behaviour may change through a full hysteresis characterisation.

Due to the time-consuming nature of the design process, a limited number of polyimide sleeves were constructed. Combined with the fragility of the sleeves and susceptibility to failure at the corners, the device was not consistently air tight. This meant that the unpressurised stiffness was characterised. This is a relatively uncommon method and due to the number of parameters related to the design of the sleeve, a direct literature comparison was not available.

Deviations from a smooth curve observed in the extension of the device were attributed to the constraint method of the sleeve. Deformation was constrained to uniaxial compression with a wooden rod inserted through the inside of the sleeve. The friction on the inside of the sleeve on the wooden rod caused moments of higher force. In the future, a new experimental rig would be designed where the securing cap attached to the distal end of the sleeve would contain longer supporting arms that could pass through the clamp at the proximal end and remove the need for an internal constraint rod.

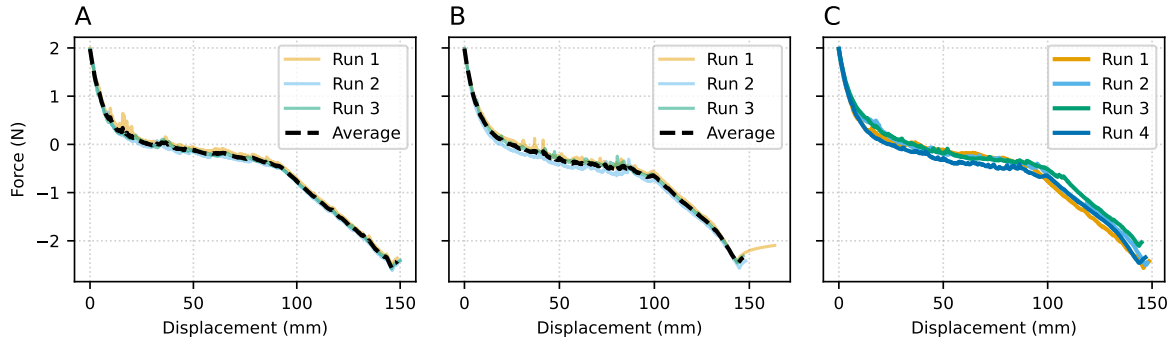

Figure 4.5: A) The initial profile of the actuator before heating. B) The profile after 3 cycles in the oven. C) A plot comparing the average force vs. displacement across all cycles

In the future, a control platform will be developed. This would include linear actuators to control the nitinol tendons and a vacuum/pressure pump to control the locomotion. This platform should allow for the non-linear relationship between the length of the device and the extending force applied by the polyimide sleeve. The unpressurised characterisation shown in this experiment sets a low bound on the stiffness of the device, when pressurised, the device will exert more force axially.

Temperature Stability of the Bellows

Temperature stability of the polyimide sleeve was evaluated through repeated exposure to high temperature (200°C, 1 hour) in an oven. This was completed on the same experimental rig as the room temperature stiffness characterisation. After 3 repetitions of heating the sleeve for an hour, it was determined that there was minimal characteristic change in the performance of the actuator after exposure to heat (Shown in 4.5). The maximum range in the stiffness of the unheated sleeve and the actuator that had been heated for 3 hours was 0.09N and the standard deviation was 0.44N. Literature results confirm the stability of polyimide to temperature, with some examples showing a total resistance to 275°C for over a year[69] and 30,000 cycles between dry ice (-79 °C) and room temperature.[74] Kapton®, the specific polyimide used here, has a thermal decomposition point at 800 °C,[69] hence the behaviour in this construction was as expected. The lack of

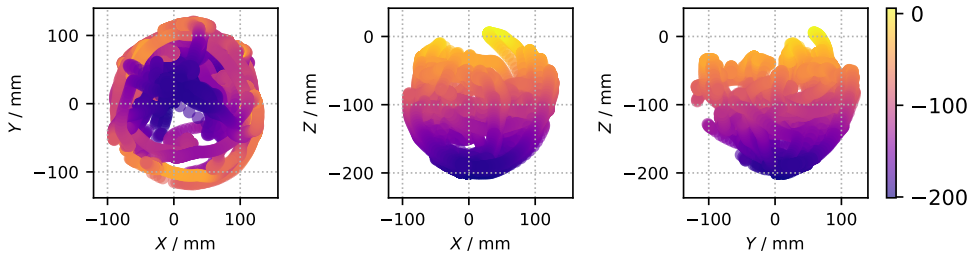

Figure 4.6: Colour maps showing the total reachable work space of a silicon continuum actuator and a protective polyimide sleeve. Colour indicates the Z displacement from resting actuator.

behavioural change implies that the assembly would be well suited to a thermally harsh environment. In the future, the heat treatment along with other degradation techniques should be examined. In the use case of space robotics, an analysis of the UV resistance whilst heating would confirm the suitability of the sleeve.

4.2.2 Workspace Characterisation

The workspace of a device is the set of all reachable tip positions that the device can achieve.[83]The resultant workspace reachable by the tip of this actuator is shown in figure 4.6. It was observed that the reachable workspace of the device in the XY plane was approximately circular, however there was a slight asymmetry in the circle, with an area of high density (associated with the relaxed actuator) at a slight positive Y shift. This plot was centred on the lowest point in the Z plane which therefore indicates a slight tip bend during the actuators relaxed state. In the X-Z view, a roughly circular workspace was observed, with the actuator achieving greater than a hemisphere, which demonstrates the ability of the actuator to extend beyond the base of the device when fully actuated. The Y-Z view shows a more triangular shape, indicating that the device doesn't have a perfectly symmetrical workspace. This behaviour was attributed to the internal friction in the tendon channels causing the device to exhibit uneven behaviour.

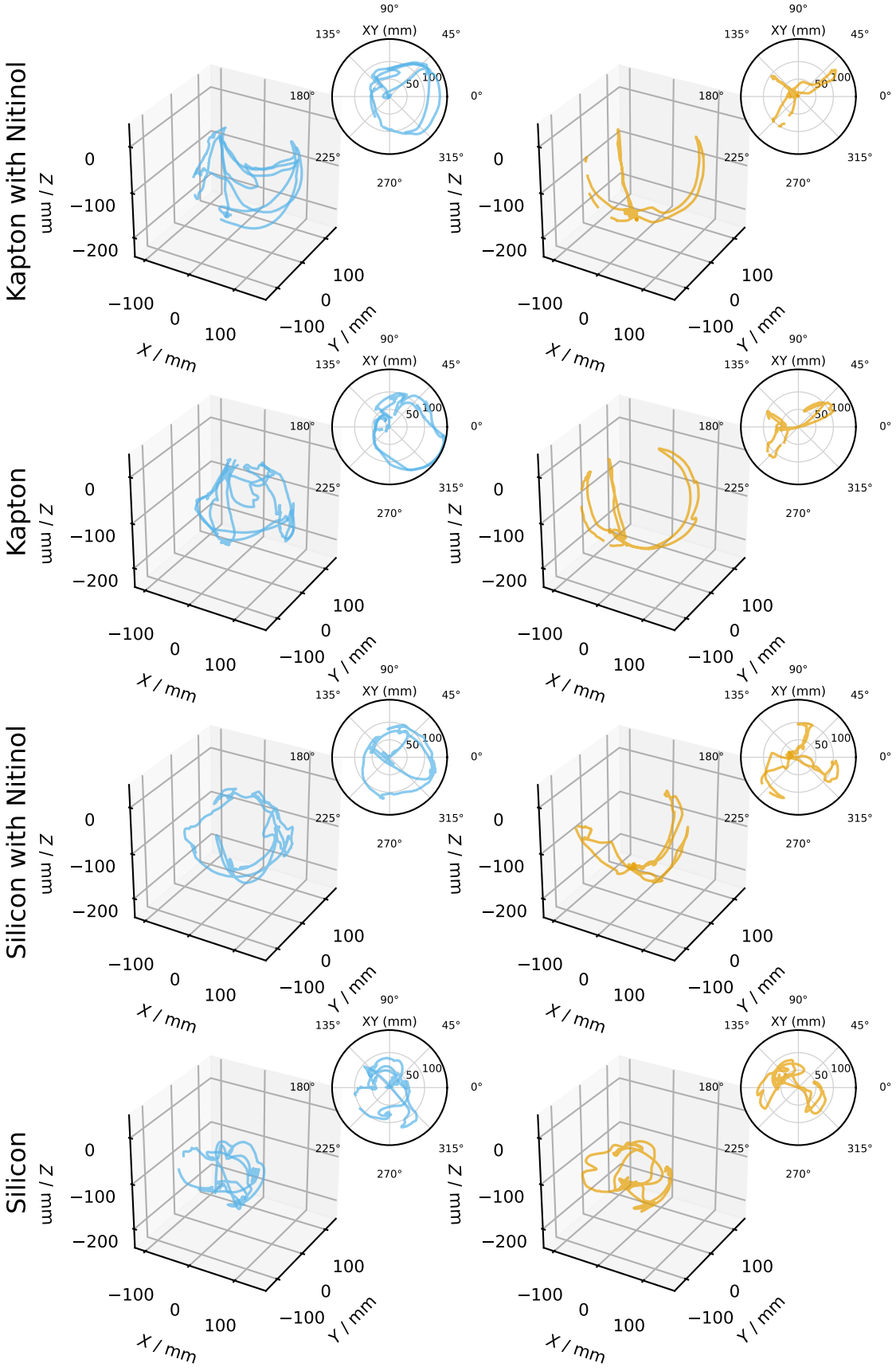

Figure 4.7: 3D and top down polar plots of the constructed continuum robot with assemblies of varying stiffness.

Uniaxial Trajectory Characterisation

Primary evaluation of the actuator performance was conducted through independent actuation of each tendon and monitoring the tip location in 3D space. Rotation matrices and XYZ shifts were applied to the data to rotate the plots and align the base of the actuator at the origin ($(x, y, z) = (0, 0, 0)$). This was tested through 4 different assemblies of increasing stiffness, labelled *Silicon* for just the silicon actuator, *Silicon with Nitinol* for the silicon actuator with NiTi (0.6mm) inserted through the central channel, *Kapton* for the silicon actuator with the Kapton® sleeve and finally *Kapton with Nitinol* as a combination of the *Silicon with Nitinol* surrounded with the polyimide sleeve. These constructions were chosen to gradually increase the stiffness of the device in an attempt to improve the controllability. It was observed that the variation in paths was lowered through increasing the stiffness, with the polyimide sleeve constraining torsional movement. Variable stiffness devices have been explored in the literature in order to utilise the advantages of soft bodies but with an improvement in dexterity through stiffening.[84,85]

Starting with the least stiff assembly, the *Silicon* actuator only, it was observed that the device was difficult to control through independent actuation of the tendons, this was presented in the plots in figure 4.7. After increasing the stiffness through the insertion of the nitinol wire in *Silicon with Nitinol*, a more controllable device was observed. The polar insert plots 4.7 showed that the trajectories were 120° apart, however also observed was a twisting motion at the tip of the device, giving the plots a "hooked" shape. This "hooking" deformation explained the reduction in vertical displacement of the actuator (see 3D plot 4.7). When adding the polyimide sleeve to constrain the torsional deformation of the actuator in the *Kapton* experiment, the "hooked" shape was minimised, however the device did not achieve trajectories with even 120° paths, instead the path exhibited a region of approximately 170° with no path. There was a small region of tracking observed within this region, however this was not linked to the other tracking points due to the actuator body partially occluding the reflective ball from view of the cameras. Again, the polar insert plots 4.7 revealed that the independent actuation of the *Kapton*

with Nitinol also contained a wide region with no tip tracking data. This implied that the polyimide sleeve was providing unwanted asymmetry which attributed to the glued seam in the polyimide sleeve.

Multiaxial Trajectory Characterisation

Another characterisation method investigated a multiaxial trajectory, whereby the first handle was pulled fully, then a second pulled fully before slowly releasing the original handle. This method was repeated with all three handles until the actuator had returned to the starting point. This method was inspired by the out-of-phase sinusoidal actuation by Chandler (2020).[86] After examining the least stiff construction, *Silicon*, it was observed that there was minimal vertical movement and a polar plot indicative of tip-twisting as observed in the previous experiment. After adding the internal nitinol, *Silicon with Nitinol* was observed to be more controllable. It was observed that the polar plot was more circular with the nitinol and had a radius of approximately 50mm. Adding the polyimide sleeve to give the *Kapton* construction resulted in an off-centre oval-shaped polar plot that had a semi-major axis of approximately 100 mm and a semi-minor axis of 75mm. Finally, the stiffest construction, *Kapton with Nitinol*, was a more regular shape in the polar plot and the greatest vertical displacement of the four constructions. The polar plot shape had a radius of approximately 100mm. As with the *Kapton* construction, the plot was not centred which was attributed to the asymmetric behaviour due to the seam in the polyimide sleeve.

Variable Length of the Actuator

Since this device has a variable length it is not limited in workspace to the hollow hemisphere in Fig.4.6. This means that in comparison to other devices without this capability the workspace is a much greater proportion of the hemisphere. In medical engineering, this is an improvement on traditional surgical catheters that need to be retracted through

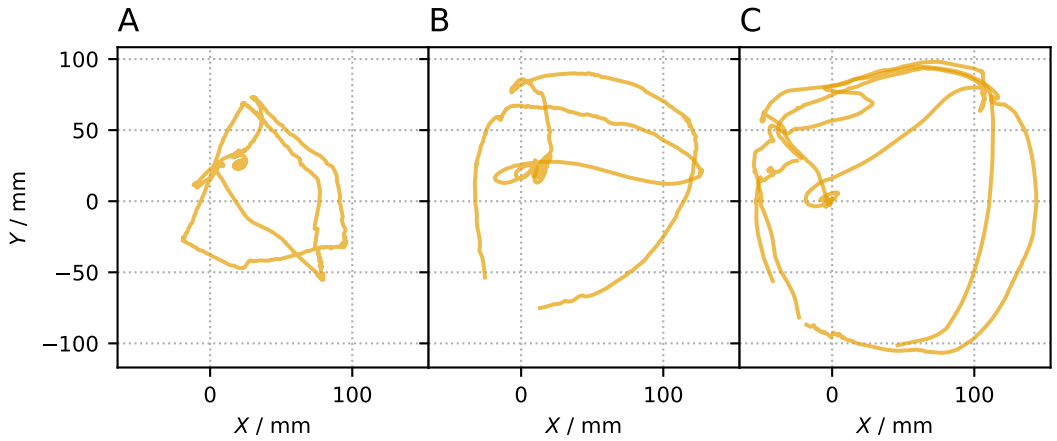

Figure 4.8: A comparison of a multiaxial trajectory of a variable length continuum robot with a polyimide sleeve and a nitinol wire as a backbone. Tested at 3 lengths: 60 mm, 90 mm and 120 mm

the patient causing large shear forces on the surrounding surfaces in the body.[4] In this polyimide-sleeved device, the retraction occurs within the body of the device which can be linked to a reduction in traumatic tissue interaction.[4]

Another workspace based experiment was conducted in which the same multiaxial actuation was used to manoeuvrer the tip of the device. This was conducted at three different lengths: 60mm, 90mm and 120mm in the *Kapton with Nitinol* construction. In the XY plane with an actuator length of 60mm, the trajectory was generally triangular with three distinct points roughly 120mm apart. As the length of the device was increased, the shape of the trajectory was observed to become more circular with a radius approximately equal to the device length. It can be observed that there was a rotation in the trajectory of the plots as the length changed. This occurred due to the cable gland which clamped the silicon actuation device to the desired length and the tightening rotation also caused a slight rotation in the actuator. Data was also collected for the other stiffness constructions, however the trends in the data were the same.

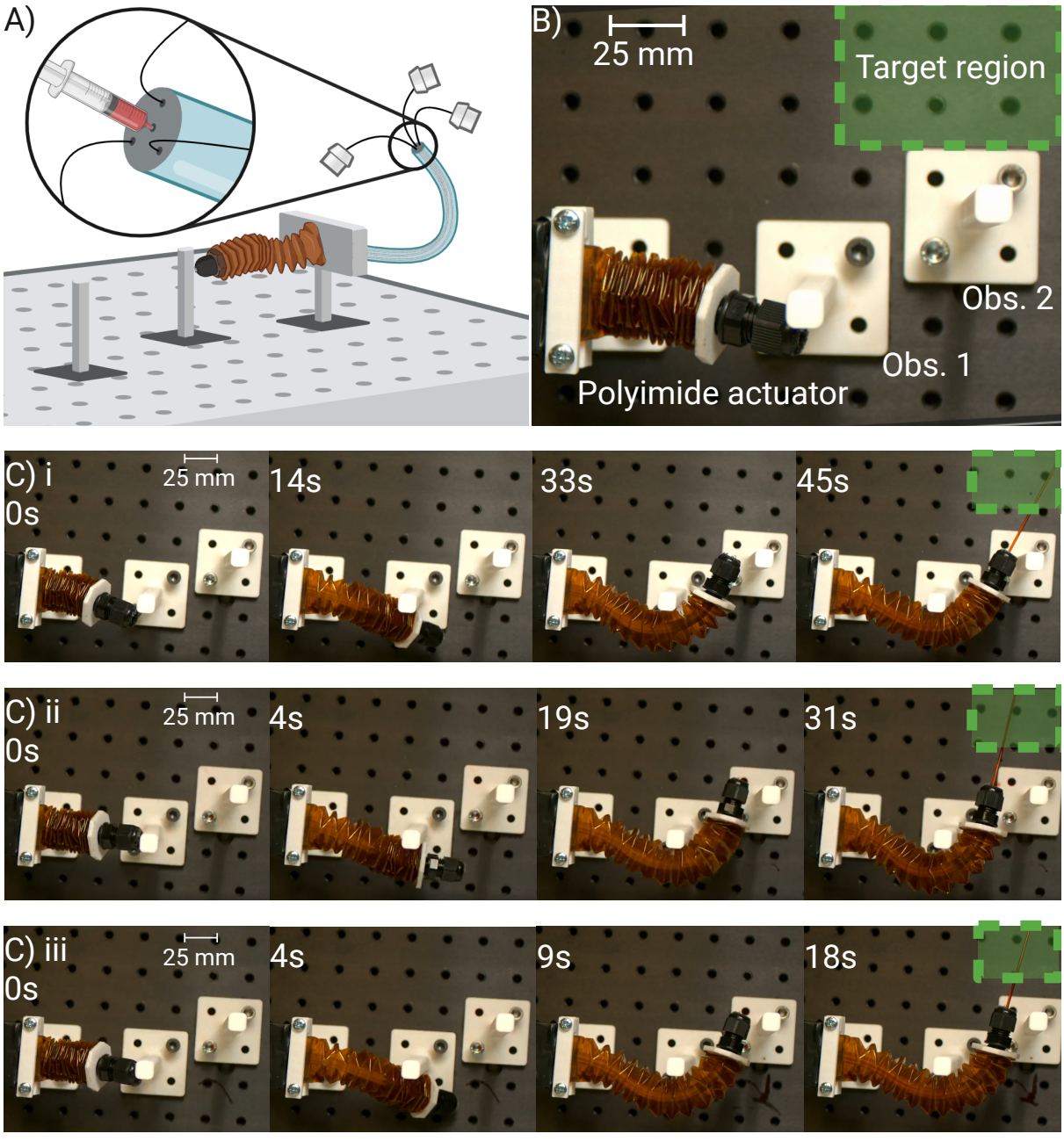

Figure 4.9: A) A cartoon of an experimental rig to demonstrate the navigation of an arbitrary route. B) a top down view of the path past two 3D printed obstacles for targeted deployment of liquid in the green marked region. C) Three repeats were completed and are marked i, ii and iii.

4.2.3 Navigation

To demonstrate the robots ability to navigate an arbitrary path, a route was set up using 3D printed obstacles. The goal of the path was to navigate to and deploy fluid within the area highlighted in green in fig. 4.9.

The start position of the path required a shorter length than the relaxed length of the actuator and therefore the operator was required to apply tension to the silicon actuator in order to retract the device before steering the device using the tendons.

The effects of the bellows elastic characteristic were noticeable, with the device easily extending to reach the relaxed length. After this point, overcoming the force of the sleeve pushing back on the silicon actuation device caused buckling internally due to the lack of rigidity of the silicon tube. Therefore the maximum path was limited to 120mm.

The introduction of a pressurised approach to controlling the locomotion of the device is beyond the scope of this. However, the addition of a pressurised controller would stop the buckling of the device at 120mm due to the internal force applied due to air-pressure rather than the silicon device. Blocked force testing of the silicon actuator would be of interest as a control method could switch between using the silicon device for extension and using air pressure to control the length.

Tip-tracking and Optimisation

An XY tip position was recorded and calculated at 0.5 second intervals, this was used to plot fig. (4.10). This plot shows the XY position of the tip of the actuator and of the two obstacles. The colour map has been added to show the time taken in seconds to complete the path. Navigation around obstacle 1 proved challenging due to the high curvature required compared to the actuator length. At around (30, -10) the noisy path was due to the 3D-printed cap on the distal end of the device colliding with obstacle 1. Across the three repeats of this navigation, this collision became less frequent through

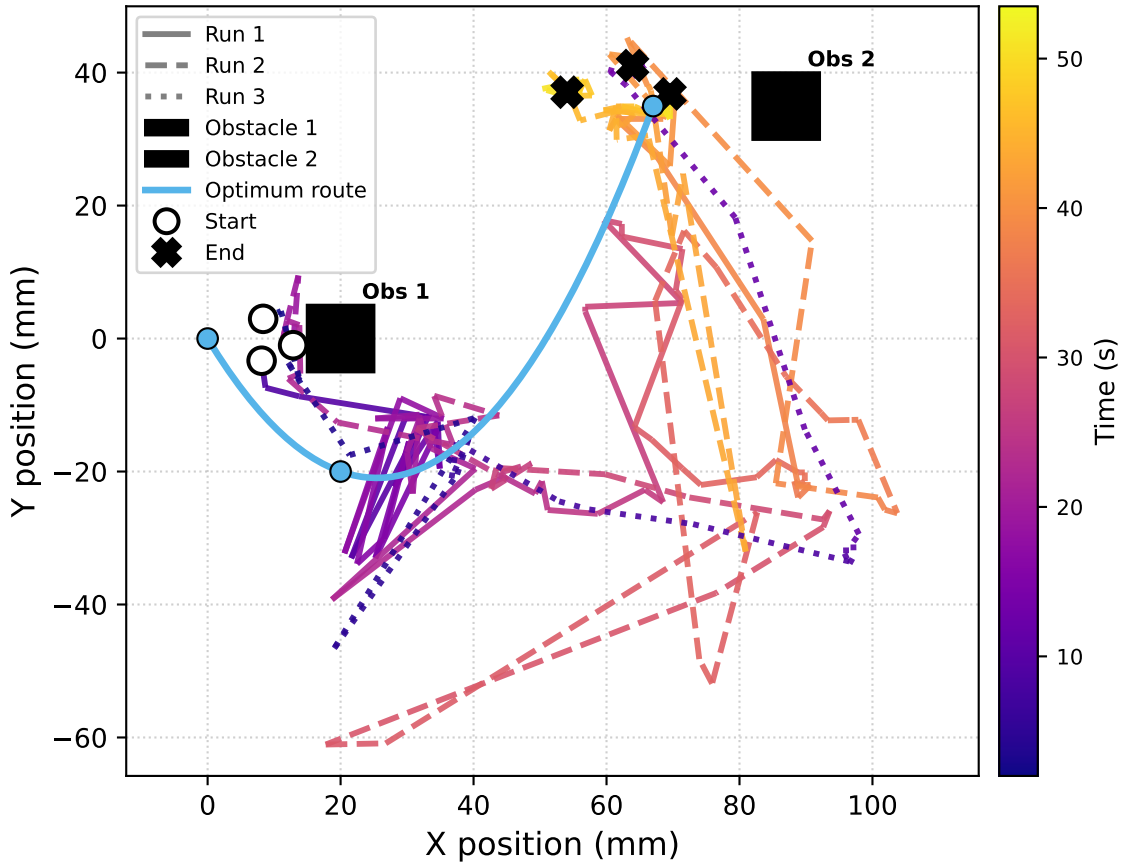

Figure 4.10: XY tip-tracking data of the arbitrary navigation past two obstacles. Time taken to complete the path is shown as the colour map. The blue line shows the optimum path through 3 points correspoding to the closest path an actuator of 30mm diameter can take to the obstacles.

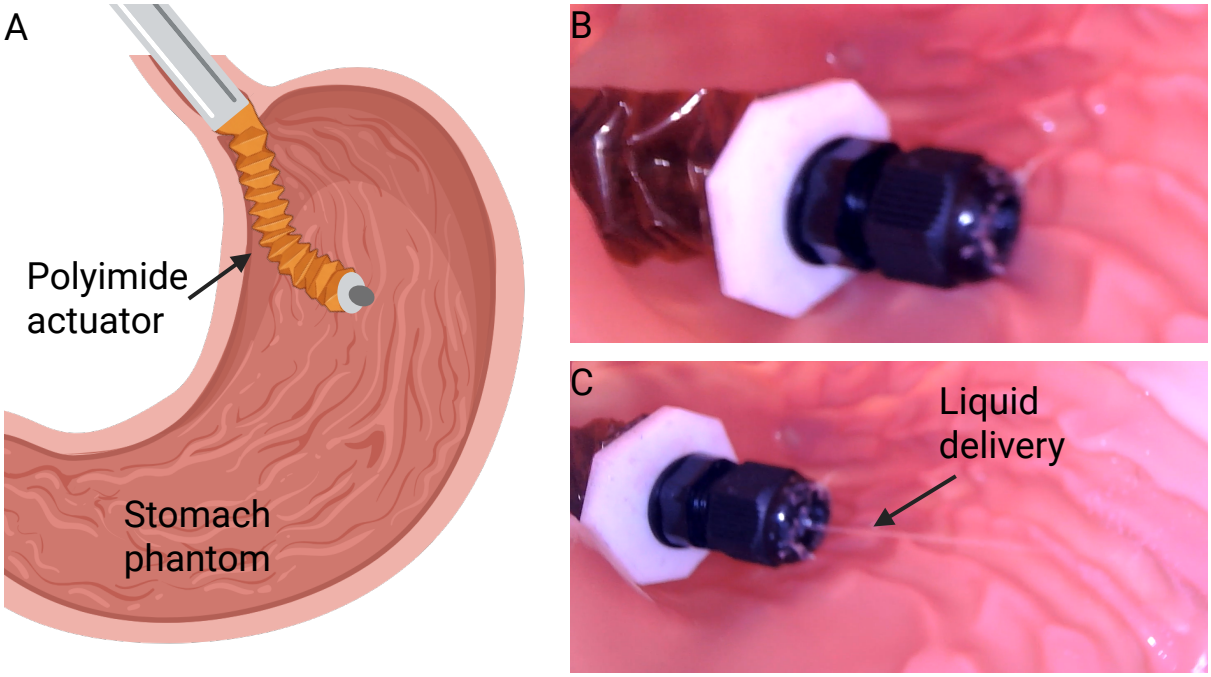

Figure 4.11: A mock-clinical demonstration of a continuum device with a polyimide protective sleeve in a stomach phantom demonstrating the ability to deploy liquid to simulate a targeted drug delivery.

user improvement of the device control. Due to the manual control, there were moments of user error in the actuation. This was observed in the rapid movement from (90, -25) to (20, -60) and then back. An optimal trajectory was obtained with the placement of three points, these correspond to the closest points that the actuator could be to the obstacles the minimise the tip angle of the device. The line connecting these points was fitted with a polynomial of degree 2 and is indicative of the optimal tip route. The true path of the actuator featured a much greater X displacement as the extension and lateral control of the device require opposite input mechanisms. Therefore a necessary duty cycling approach of axial growth and lateral actuation was adopted. In the future, the device will be tracked using the Optitrack cameras to allow Z-displacement to be calculated.

Clinical Case Study: Drug Delivery in Stomach

Finally, we demonstrated the use of the actuator in a stomach training simulator (fig. 4.11) for the application of targeted drug delivery. Here the same assembly as the previous navigations and the addition of water to simulate the drug delivery was used. An external

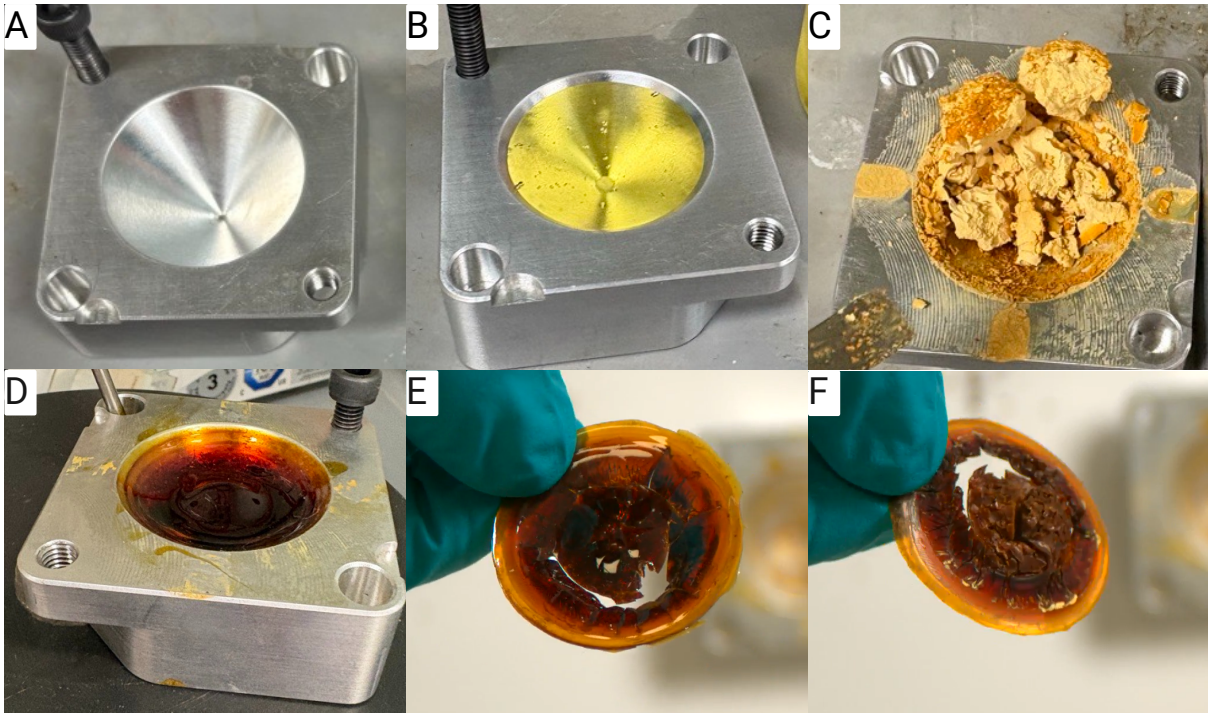

Figure 4.12: A) A conical test mould. B) Polyamic acid in NMP filling the mould. C) A failed curing of polyamic acid after heating to 150°C for 10 minutes. D) A partially successful curing of polyamic acid after drying NMP solvent for 1 week at 100°C. Sample heated to 150°C to cure. E) Resultant successful extraction of cured polyamic acid. F) Failed region at the vertex of the cone due to partial solvation of polyamic acid.

camera (camera details) was inserted through a separate entry into the stomach to give an approximate location for completing the navigations. The device was inserted into the stomach phantom at the shortest state before being extended and controlled from the section of the device outside the phantom. This demonstrated the ability of the device to reach areas and deploy liquids.

4.2.4 Moulding of Polyamic Acid

Development of a moulding process for components would increase the design versatility for robotic purposes. Polyamic acid undergoes a thermal imidisation reaction at 150-200°C to form polyimide. The formation of components other than films is desirable for the manufacturing of components needing the high performance properties of polyimide. An arbitrary moulded component was formed after drying PAA out of N-methyl Pyrollidine (NMP) solvent for 1 week.

An attempt was made by pouring PAA in NMP into a mould and heating to 150°C on a hotplate, however after 10 minutes a yellow powder was obtained. It was assumed that the lack of a unified structure was due to the solvent interrupting the curing process. Therefore a second attempt was made with the aim to remove the solvent before curing. The difficulty here was an avoidance of the thermal imidisation threshold of 150°C but still to evaporate the NMP solvent (boiling point : 202°C). Therefore the use of a Smartvap, a piece of equipment that can remove solvent at a lower temperature than literature boiling points, was necessary. It was observed that the viscosity of the PAA increased as NMP was evaporated. Therefore the in situ drying of the solvent was necessary, however the Smartvap was designed to fit a vial with an internal diameter of around 12mm, this motivated the design of a 3D printed component to connect the Smartvap to the mould. Therefore an adaptor was designed in Fusion360 and printed in TPU to allow the Smartvap to connect to the top of the mould to allow the PAA to be dried in situ. After a week of drying, the TPU adaptor was removed and the mould was placed onto a hot plate set to 150°C. After 1 hour, the mould was cooled, removed and the moulded component was removed from the mould with a scalpel. The base of the cone had a smooth texture whereas at the tip, one single piece was not obtained. This was again assumed to be due to the PAA partially being in solution.

4.3 Experimental Methods

4.3.1 Stiffness Characterisation

To characterise the stiffness profile, a linear stage (THORLABS Linear Translation Stage 150mm) was used with a load cell (900 series S Beam, LoadCellShop) held in a clamp (SmallRig Magic Arm). This was attached to the end of the polyimide sleeve which was clamped in a 3D printed stand. The whole experimental setup was attached to a breadboard (THORLABS). The linear stage was set to the resting length of the actuator

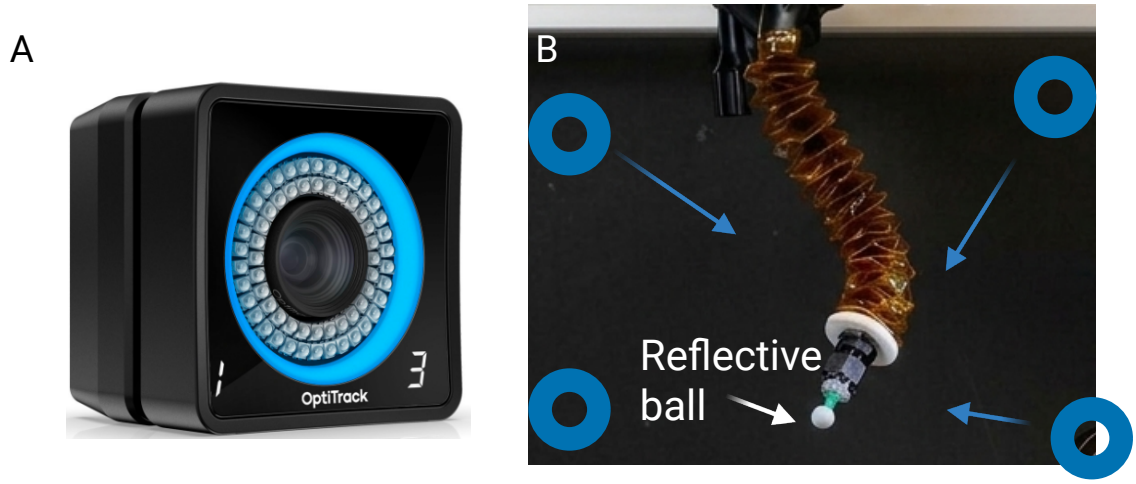

Figure 4.13: A) An Optitrack camera. B) Relative locations of Optitrack cameras and their view shown in blue to characterise the workspace of the actuator

(120mm) and the load cell was tared. The linear stage was set to compress the actuator to 30mm this corresponded to 0mm on the linear stage. The load cell was set to record and the linear stage was controlled to 150mm (sleeve length 180mm) at a speed of 0.5mm s^{-1}. The speed of the actuator was translated into the displacement using: $v = \frac{d}{t}$. The load cell data was collected using DSCUSB Toolkit and processed in Python.

Temperature Stability of the Polyimide Sleeve

Using the same experimental set-up as the stiffness characterisation. The polyimide sleeve was placed in an oven at 200°C for 1 hour, left to cool and then compressed and stretched as before. This was repeated 3 times before data was proccessed and plotted in Python.

4.3.2 Workspace Characterisation

To characterise the usable workspace of the device, an experiment was set-up with the four camera Optitrack system and Motive 2.3 software to track the tip position of the device in 3D space with submillimeter accuracy (mean error: 0.0103mm). Two trajectories were examined, a uniaxial characterisation with the actuation of each tendon separately and a multiaxial characterisation through cyclical actuation of each tendon. This was

completed for 4 different assemblies of actuator, the silicon actuator alone, the silicon actuator with the polyimide sleeve and both of these with a NiTi (0.6mm diameter) wire inserted through the central lumen. This was completed for three different lengths of actuator, 60mm, 90mm and 120mm. The 3D-printed tip with cable gland was attached to the end of the device to hold a retroflective ball that the Optitrack cameras could track. A size exclusion filter was set to only track the tip-mounted ball. Data was processed and plotted in Python

4.3.3 Moulding of Polyamic Acid

Polyamic acid in N-methyl pyrollidone (15-16 wt. %, Sigma Aldrich) was poured into a conical mould and heated to 150°C until thermal imidisation occured. This was attempted without removing the N-methyl pyrollidone (NMP) from the sample and also attmepted after drying the sample for 1 week with a SmartVap. Without removing NMP, the heating caused a yellow powder to be formed after 20 minutes. The dried sample formed a continous solid layer after 1 hour of heating. The solid layer occured at the circular base of the cone and broke into pieces (approx. $2mm^3$) at the vertex.

Chapter 5

Conclusion

This report showed initial steps towards the development of a self healing polyimide, this was attempted through hydrolysis of a polyimide to a precursor, polyamic acid, and subsequent crosslink formation between chains of this precursor. The success of hydrolysis attempts were minimal, attributed here to a low concentration of alkaline reagent used. Crosslinking of polyamic acid was also unsuccessful. Here this was attributed to an intramolecular imide formation blocking the attachment of a crosslinking reagent. An unexpected result discovered during catalytic crosslink attempts was a reduction in temperature required for thermal imidisation of polyamic acid to polyimide. This has potential to provide a greener production process of polyimide due to a 40°C decrease in temperature.

Other experiments performed with the same material had a soft-robotic focus. This involved the design and construction of an actuator shown to withstand temperatures of 200°C with no loss in capability. This device had a greater than hemispherical workspace with a maximum radius of 120mm. Follow-the-leader navigation with this device was demonstrated to 120 mm achieving a maximum tip angle of 74°. Additional capabilities of this device were also shown such as targeted drug delivery in a stomach based simulation.

In order to improve the hydrolysis of polyimide, a greater concentration of base reagent

should be used. To achieve similar results to other sources, a concentration between 1M and 5M is required. Crosslinking improvements are less trivial due to the highly competitive intramolecular imide reformation. Hindering this reaction through the addition of nitrogen protecting groups is an option. However, avoiding the imide ring completely is also an option by adding additional substituents to aromatic rings contained within the oxydiphenylene section of the polyimide structure.

The design of the polyimide sleeve can also be updated, altering the angle of the Accordion fold pattern to 45° has been shown in other literature results to be the most compressible design. Minimising the axial stiffness whilst maintaining the torsional rigidity creates the optimal design. Once the design has been improved, the development of a control platform including pressurised locomotion and linearly actuated tendons will open possibilities for modelling and more precise manoeuvrability.

In the future, the chemical contributions to the robotics field will become more apparent. When crosslinking attempts become successful, a shift in reagents can make way for *dynamic* Diels-Alder groups in order to re-bind chains that have been separated due to damage. This highly functional material can replace the current protective sleeve shown in this work and extend the lifetime of soft-robotic actuators.

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Appendix A

1H NMR plots

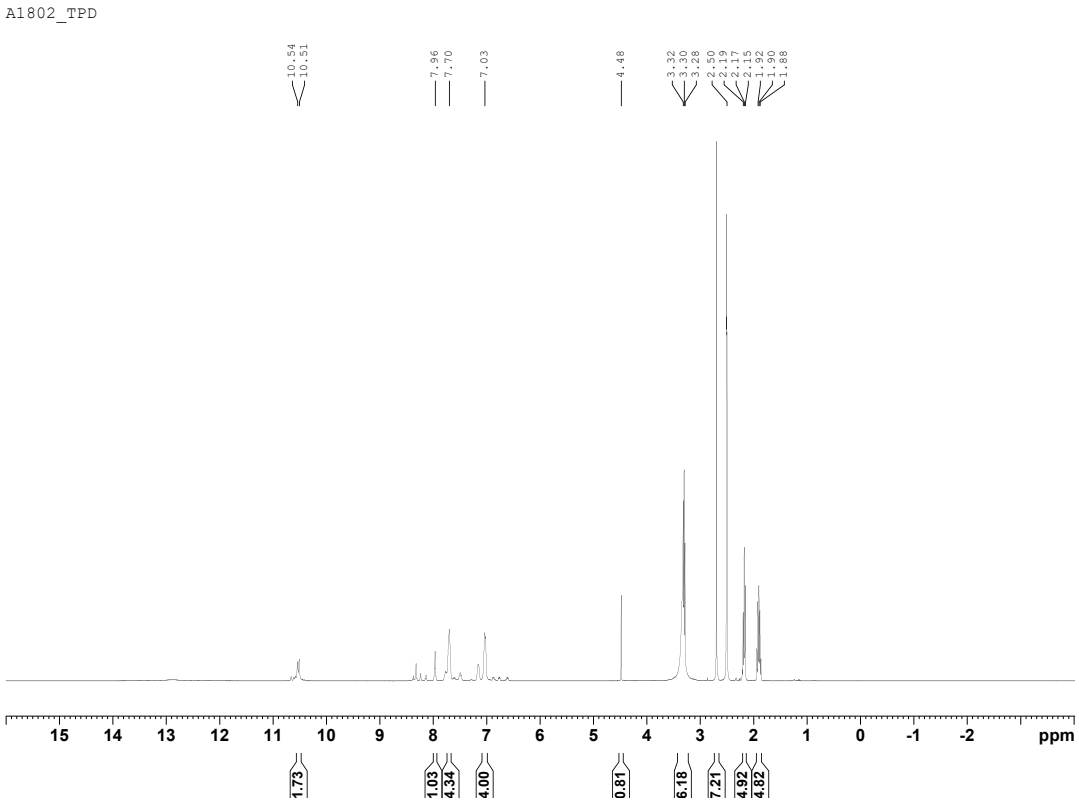

Figure A.1: ^{1}H NMR plot of attempted crosslinked sample with $ZnCl_2$ and ethylene carbonate.

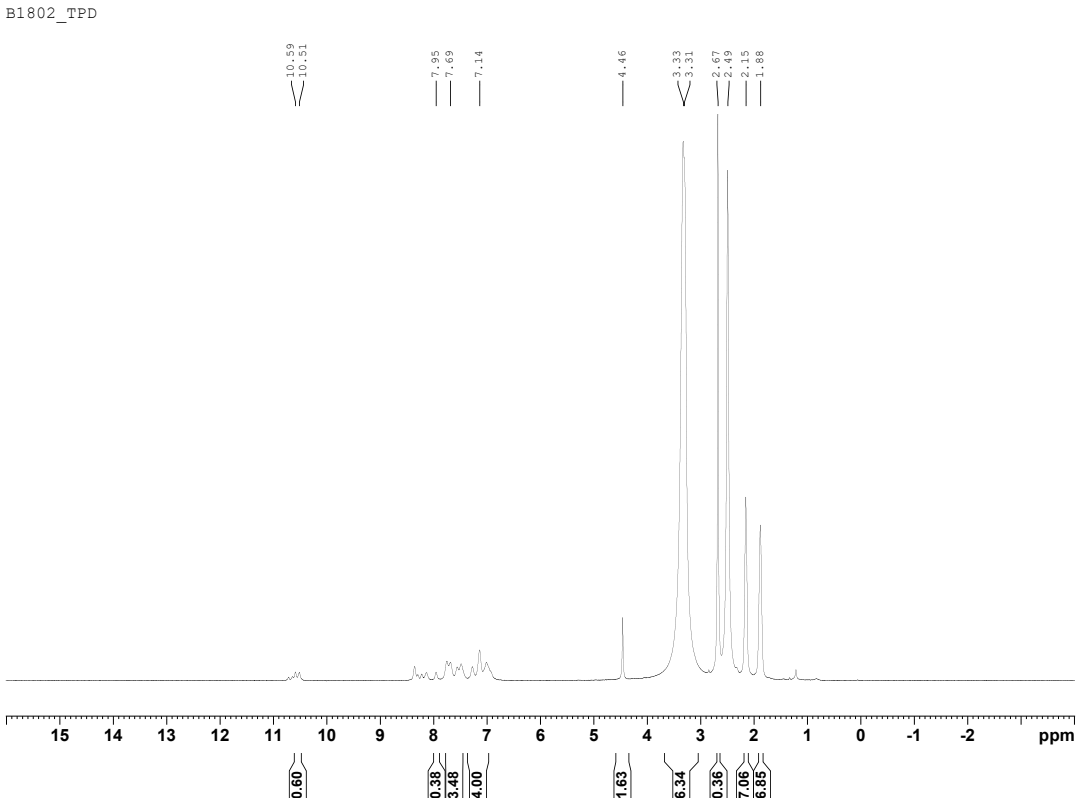

Figure A.2: ^{1}H NMR plot of attempted crosslinked sample with $FeCl_3$ and ethylene carbonate.

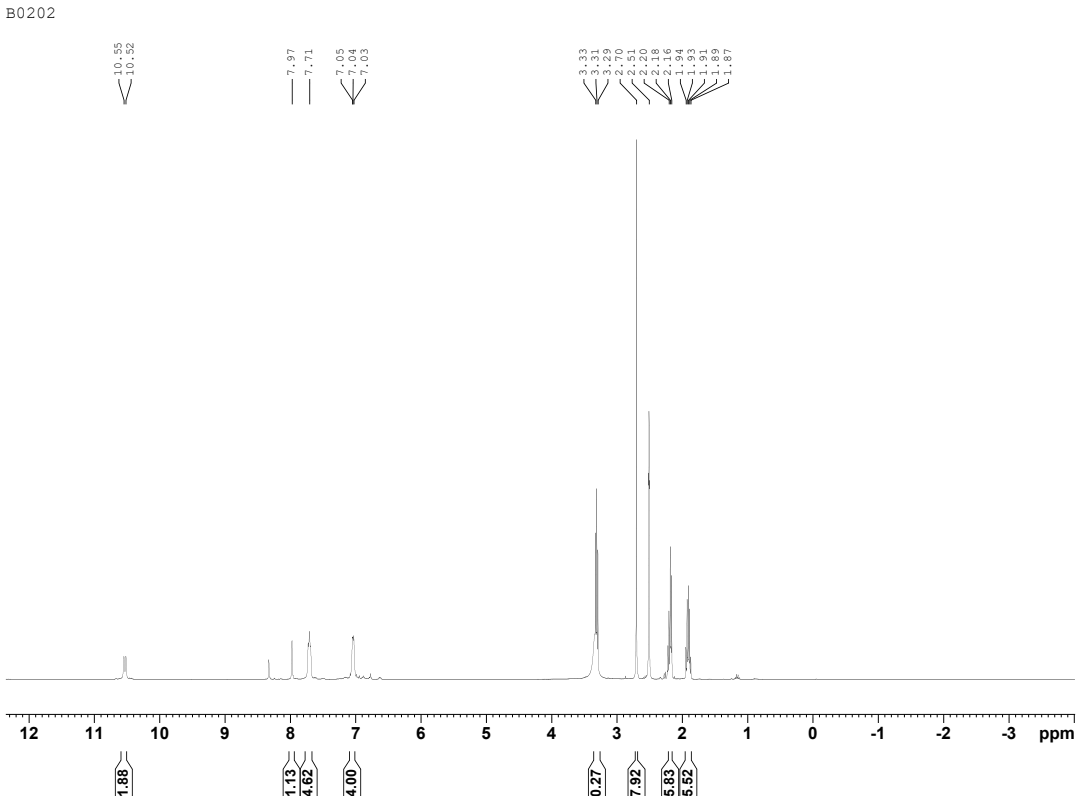

Figure A.3: ^{1}H NMR plot of polyamic acid